

Nearly Optimal Community Recovery by Spectral Likelihood Decoding

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Abstract

Spectral clustering compresses a network into a small number of eigenpairs, whereas likelihood-ratio classification uses block-dependent logarithmic contrasts. We prove that a decoder using only the retained adjacency eigenpairs attains the leading Chernoff error exponent for every fixed community order K . The stochastic block model has positive community proportions, comparable positive block probabilities, a uniformly conditioned K -dimensional signal, and minimum expected degree $\Omega(\log n)$. Degrees may differ across communities and grow faster than $\log n$; the block matrix may vary with n . Our sparse working score assumes that its largest entry tends to zero. The procedure starts at the global reconstructed profile mean, grows components by their all-node likelihood gain, and jointly refits after each addition. It then repeats global-gain component replacements with uniformly initialized proposal weights until no likelihood gain reaches an explicit vanishing relative threshold. This deterministic stopping certificate ensures coverage of all K true profiles and weakly consistent responsibilities. One final EM update and likelihood-ratio decoding give $\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}$, without an assumed weak initializer. The proof controls the dependent full-graph spectral reconstruction at the information-scale failure probability and establishes an invariant local EM basin. Archived experiments on 240 graphs evaluate the growing branch and a historical single replacement pass; the repeated certified procedure has separate implementation checks.

1 Introduction

Community recovery in the stochastic block model (SBM) is a basic problem in network statistics. The model assigns each node to an unknown community and makes edge probabilities depend on the communities of their endpoints [6]. Spectral methods reduce the adjacency matrix to a small number of eigenvectors, whereas likelihood methods compare how well alternative community assignments explain the observed connections. We show that likelihood-ratio information can be decoded from the retained spectrum to achieve nearly optimal community recovery for every fixed number $K \geq 2$ of communities. Our theorem allows unequal expected degrees and a range of sparse densities, with minimum expected degree at least a constant times $\log n$.

This question concerns the decoding rule as well as the embedding. The oracle classifier for an SBM weights connections by logarithms of block probabilities and includes the contribution of absent edges. For a general connectivity matrix and unequal community sizes, its decision boundaries need not agree with nearest-center boundaries in the spectral embedding. Nevertheless, a projection preserves every linear statistic whose weight vector belongs to the projected subspace. This gives a route from spectral compression to a likelihood decoder.

Optimal recovery and likelihood refinement. The information-theoretic limits of the SBM are well developed. Abbe and Sandon [1] characterize exact recovery in the general logarithmic-density SBM through Chernoff–Hellinger divergence and give efficient algorithms reaching that threshold. Zhou and Li [13] study the optimal error rate through Chernoff bounds and its application to community detection. We use these works to identify the oracle benchmark, while the statistical objective of the present paper is the leading-exponent form

$$\exp\{-(1 + o(1))I\},$$

where I is the relevant pairwise Chernoff information. Throughout this paper, *nearly optimal* means attaining the leading oracle Chernoff exponent. We do not claim the refined multiplicative rate or a new global minimax lower bound.

Spectral initialization followed by likelihood refinement is also an established approach. Amini, Chen, Bickel and Levina [3] fit mixtures to blockwise connection counts using pseudo-likelihood and iterative updates. Gao, Ma, Zhang and Zhou [5] prove that a weakly consistent initializer followed by penalized local likelihood refinement achieves optimal misclassification performance under their assumptions. Zhou and Amini [12] obtain optimal bipartite recovery using spectral initialization and pseudo-likelihood classification, and analyze a broader family of EM-type biclustering updates. A recent preprint of Wang [9] analyzes spectrally initialized variational EM in a general SBM, obtaining a leading Chernoff exponent while estimating parameters and reusing the original graph. In these methods the refinement uses information calculated from the original edge observations. Once our selected adjacency eigenpairs have been computed, initialization, parameter updates, classification and component replacement use only those eigenpairs and quantities reconstructed from them.

Likelihood information in spectral representations. Spectral optimality is known in important special cases. Abbe, Fan, Wang and Zhong [2] establish a first-order entrywise approximation for eigenvectors and show that, in the equal-sized symmetric two-community SBM at logarithmic density, an untrimmed spectral method attains the optimal recovery threshold and misclassification exponent. Their result already demonstrates that a spectral representation can retain the decisive local evidence. For homogeneous planted-partition SBMs with multiple, possibly unequal communities, Zhang [10] gives matching error exponents for high-degree trimming followed by eigenvalue-weighted spectral clustering and K -means. His spectral exponent is distinguished from the information-theoretic exponent, so a sharp analysis of a spectral algorithm does not automatically establish its statistical optimality.

Probabilistic inference based on spectral embeddings predates the present work. Suwan et al. [8] use Gaussian-mixture limits of adjacency spectral embedding to construct an empirical Bayes prior for block membership inference. This is a relevant example of using distributional information beyond Euclidean clustering; it does not establish a sparse large-deviation equivalence between our spectral scores and the oracle edge likelihood ratio. For lower densities, Zhou and Amini [11] provide data-driven adjacency regularization and consistency analysis in the general bipartite setting. These consistency results do not by themselves prove preservation of the oracle error exponent.

Our approach. We retain the K eigenpairs of the adjacency matrix with largest absolute eigenvalues, including signal eigenvalues of either sign, and form positive reconstructed connection profiles q_i . The positivity floor is proportional to the observed spectral scale divided by n , so it adapts to the connection probabilities. A sparse likelihood-based working score compares these

profiles with candidate community profiles. Its pairwise difference has the algebraic form of a likelihood ratio.

Initialization starts a single component at the global profile mean, adds a node-profile candidate according to its gain over all nodes, and refits every component after each addition. We then iterate a deterministic component-replacement step. Each round compares a refit of the current profiles with trials obtained by removing one component, adding the node profile with largest gain relative to the remaining components, and jointly refitting. Each trial starts with uniform weights. An improvement above the stated threshold is accepted, and the process repeats until the threshold is not met. A final EM update precedes decoding. This is an explicit iterative extension of the growing procedure. All candidate choices, fits and acceptance decisions use the same spectral profiles and working likelihood.

The general- K guarantee rests on the replacement stopping certificate. If the fitted profiles leave a population community poorly represented, then with K fitted components and K separated population profiles, another fitted component can be removed at a sufficiently small loss. The all-node gain supplies a candidate for the omitted community, and refitting cannot reduce the trial's likelihood. Repeated accepted improvements are bounded by the available objective range. At termination, the certificate places the fitted responsibilities in a weak-recovery basin, after which local EM analysis gives the oracle error exponent. The proof does not impose weak recovery on an unspecified initializer.

The spectral interpretation has a precise algebraic basis. If $\hat{\Pi}$ denotes the projector onto the retained empirical eigenspace, its signed rank- K reconstruction equals $A\hat{\Pi}$. With a population SBM projector, every block-constant likelihood-ratio contrast is preserved when the block matrix is full rank. Empirical projection, positivity clipping and estimated parameters introduce additional errors. We control those errors for the whole-graph empirical profiles through first-order entrywise eigenspace analysis and deterministic likelihood and EM bounds. The reconstructed profiles are dependent, noninteger quantities, so the numerical fitting objective remains a working likelihood. Its relation to the oracle Bernoulli likelihood is established by the statistical analysis.

The paper makes three contributions. First, it gives a deterministic spectral-only likelihood decoder with a complete initialization and refinement rule for every fixed $K \geq 2$. Second, Theorem 3.7 proves the end-to-end bound

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq \exp\{-(1 + o(1))I_n\},$$

where I_n is the minimum pairwise oracle Chernoff information. The hypotheses require positive community proportions, comparable positive block probabilities tending to zero, minimum expected degree $\Omega(\log n)$, and a uniformly nondegenerate rank- K signal. They do not fix the normalized block matrix or require equal expected degrees. The bound implies exact recovery when $\liminf I_n / \log n > 1$. The local EM theorem and algebraic preservation results are tools for this complete recovery theorem.

Third, paired experiments on 240 graphs evaluate the original growing rule and a historical likelihood-accepted parameter-replacement procedure. They diagnose severe local optima in which nearby communities share one mixed profile while another community receives two components, and show how all-node likelihood gains or accepted replacements repair the observed failures. These archived results concern the procedures actually evaluated. They do not constitute a benchmark of the newly specified repeated-replacement Algorithm 1. Fresh checks of that algorithm verify its implementation on models with more than two communities; they do not estimate an asymptotic error exponent.

Section 2 specifies the model, spectral scores and algorithm. Section 3 gives the recovery results. Section 4 presents experiments and failure diagnostics, and Section 5 concludes. Proofs appear in the appendix.

2 Spectral likelihood-ratio decoding

2.1 Model and retained information

Let A be the adjacency matrix of an undirected graph without self-loops. Conditional on community labels $z = (z_1, \dots, z_n) \in [K]^n$, its upper-triangular entries are independent and

$$A_{ij} \sim \text{Bernoulli}(P_{z_i z_j}), \quad i < j, \quad A_{ji} = A_{ij}, \quad A_{ii} = 0. \quad (2.1)$$

We follow the SBM notation of Zhou and Li [13]: $P = P_n \in (0, 1)^{K \times K}$ is the symmetric block probability matrix, $p_{aj} = P_{a, z_j}$, and p_{a*} is the connection profile of community a . The known number $K \geq 2$ is any fixed integer. Write $Z_{ia} = \mathbf{1}\{z_i = a\}$, $N = Z^\top Z = \text{diag}(n_a)$, and $n_a/n \rightarrow \pi_a > 0$. Define $P_{\min} = \min_{a,b} P_{ab}$ and $P_{\max} = \max_{a,b} P_{ab}$. The main results assume, for fixed positive constants c_r, c_s, c_d ,

$$P_{\min} \geq c_r P_{\max}, \quad P_{\max} \rightarrow 0, \quad \sigma_{\min}(N^{1/2} P N^{1/2}) \geq c_s n P_{\max}, \quad (2.2)$$

and

$$\min_i \mathbb{E}[\deg(i) \mid z] \geq c_d \log n, \quad \mathbb{E}[\deg(i) \mid z] = \sum_{b=1}^K (n_b - \mathbf{1}\{z_i = b\}) P_{z_i b}. \quad (2.3)$$

The density condition is a lower bound. Expected degrees may differ across communities and may grow faster than $\log n$; neither P nor its normalized shape is fixed. The conditioning assumption in (2.2) supplies the signal separation needed by the retained K -dimensional representation. It permits negative signal eigenvalues.

The $n \times n$ mean matrix differs from the rank- K matrix $Z P Z^\top$ only on its diagonal:

$$\mathbb{E}[A \mid z] = Z P Z^\top - \text{diag}(P_{z_i z_i}). \quad (2.4)$$

The small diagonal correction matters for exact identities. The representation consists of the K empirical eigenpairs of A with largest absolute eigenvalues:

$$AU = U\Lambda, \quad U^\top U = I_K, \quad \tilde{A} = U\Lambda U^\top. \quad (2.5)$$

All subsequent fitting accesses only (U, Λ) . This restriction excludes an additional edge-based refinement. It also excludes computing separate leave-one-out eigendecompositions as an undeclared algorithmic step. The representation is computed from the original adjacency matrix.

2.2 Spectral profile scores

A signed low-rank reconstruction can contain nonpositive entries. To define the logarithms used in candidate scoring, set

$$q_{ij} = \max\{\tilde{A}_{ij}, \varepsilon_n\}, \quad \varepsilon_n = \frac{c_0}{n} \max\{\|\Lambda\|_{\text{op}}, 1\}, \quad 0 < c_0 < c_r/2. \quad (2.6)$$

The floor is computed from the retained eigenvalues. On the spectral event in Lemma 3.5, it is a fixed positive fraction of $P_{\max}$ and lies below all true block probabilities. It makes the profile logarithms well defined. The adjacency eigendecomposition in (2.5) uses A as given. The reconstruction retains its diagonal. For a candidate profile $\hat{p} \in (0, \infty)^n$, define

$$\ell_i(\hat{p}) = \sum_{j=1}^n q_{ij} \log \hat{p}_j - \sum_{j=1}^n \hat{p}_j. \quad (2.7)$$

The score is motivated by the first-order sparse expansion of a Bernoulli log likelihood: $\log(1-p) = -p + O(p^2)$. Lemma 3.2 controls the resulting pairwise approximation. Because q is a dependent spectral reconstruction, (2.7) defines a working fitting criterion. It is not the exact likelihood of the original graph. Terms depending only on q_i do not affect candidate comparisons or fitting.

For profiles $\hat{p}_1, \dots, \hat{p}_K$ and weights $\hat{\pi} \in \Delta_K$, the working mixture objective is

$$\mathcal{L}(\hat{\pi}, \hat{p}) = \sum_{i=1}^n \log \left(\sum_{a=1}^K \hat{\pi}_a e^{\ell_i(\hat{p}_a)} \right). \quad (2.8)$$

The final label is

$$\hat{z}_i = \arg \max_{a \in [K]} \{ \ell_i(\hat{p}_a) + \log \hat{\pi}_a \}. \quad (2.9)$$

Thus classification uses pairwise working likelihood ratios, with estimated prior weights determining the thresholds. We evaluate the permutation-invariant mislabeled fraction

$$\text{Mis}(\hat{z}, z) = \min_{\sigma \in \mathfrak{S}_K} \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\sigma(\hat{z}_i) \neq z_i\}, \quad (2.10)$$

where $\mathfrak{S}_K$ is the set of label permutations. The fitted profiles $\hat{p}_{a*}$ are responsibility-weighted means of the observed profiles; the implementation does not impose the exact SBM constraint $\hat{p}_{aj} = P_{a,z_j}$. Although these profiles have n coordinates, every M-step keeps them in the row convex hull of q ; Proposition B.4 records this restriction.

2.3 Global-mean start and growing likelihood gains

The procedure starts with the global profile mean,

$$\hat{p}_1^{(1)} = \frac{1}{n} \sum_{i=1}^n q_i, \quad \hat{\pi}_1^{(1)} = 1. \quad (2.11)$$

This is the fitted one-component solution, rather than a node chosen using a top-fraction subset. It requires no subset-size parameter and removes no observations.

At stage $t = 2, \dots, K$, let $\hat{p}_1^{(t-1)}, \dots, \hat{p}_{t-1}^{(t-1)}$ be the profiles fitted at the preceding stage and set $h_i^{(t-1)} = \max_{a < t} \ell_i(\hat{p}_a^{(t-1)})$. For a node candidate c not previously added, define

$$\Delta_t(c) = \sum_{i=1}^n (\ell_i(q_c) - h_i^{(t-1)})_+, \quad c_t \in \arg \max_c \Delta_t(c). \quad (2.12)$$

Append q_{c_t} to the current $t-1$ fitted profiles, initialize the t weights uniformly, and fit all t profiles jointly on all n observations by the updates below. The fitted endpoint supplies the next stage's

profiles. A global mean counts as the first component, so only $K - 1$ node candidates are added. Previously chosen node IDs are excluded from subsequent candidate scans, and ties are broken deterministically.

The all-node gain rewards profiles that improve the fit to observations. The growing endpoint initializes the following fixed- K replacement procedure. Its guarantee is established by a stopping certificate for the common objective.

2.4 Repeated global-gain replacement

Define the profile divergence

$$D(x\|y) = \sum_j \{x_j \log(x_j/y_j) - x_j + y_j\}, \quad T_n = \sum_{i,j} q_{ij}. \quad (2.13)$$

The divergence equals the score loss $\ell_x(x) - \ell_x(y)$. Starting at the K -component growing endpoint, evaluate these proposals in each round:

1. Refit the current profiles with weights initialized at $1/K$.
2. For each component r , remove it and choose a row q_{c_r} maximizing

$$\sum_i \left(\ell_i(q_c) - \max_{a \neq r} \ell_i(\hat{p}_a) \right)_+.$$

Replace profile r by q_{c_r} , initialize all weights at $1/K$, and jointly refit. The candidate scan includes all n rows, including previously selected nodes.

Each proposal may use any prescribed finite number of exact EM updates, including zero. Compare all endpoints and the current fit by the same $\mathcal{L}$. Accept the best proposal if its increase is at least $T_n/\sqrt{\log n}$, and repeat. Stop if no proposal attains that increase. For a deterministic iteration limit, stop after $\lceil (\log n)^2 \rceil$ accepted rounds. This cap is eventually inactive on the spectral event.

At the chosen endpoint, compute the E-step responsibilities and perform one additional profile and weight M-step before decoding. A zero-mass component retains its previous profile. The extra uniform-weight proposal and uniform starts for replacements are part of the certificate; they prevent fitted component weights from hiding a poorly covered population profile.

Theorem 3.7 proves this deterministic procedure for every fixed K . The key observation is that with K true profiles and K fitted components, a missing true profile can improve a one-component replacement: either its nearest component serves only that profile, or another component is unused by all nearest-profile assignments. Rejection of all global-gain replacements consequently forces coverage of every true profile. The proof controls the soft mixture objective as well as the hard profile divergence.

2.5 Alternating fitting and parameter replacement

At a stage with s components, initialize $\hat{\pi}_a = 1/s$ and perform soft updates (replace K by s in the displayed sums)

$$\tau_{ia} = \frac{\hat{\pi}_a e^{\ell_i(\hat{p}_a)}}{\sum_b \hat{\pi}_b e^{\ell_i(\hat{p}_b)}}, \quad N_a = \sum_i \tau_{ia}, \quad (2.14)$$

$$\hat{p}_{aj}^{\text{new}} = \frac{\sum_i \tau_{ia} q_{ij}}{N_a}, \quad \hat{\pi}_a^{\text{new}} = N_a/n. \quad (2.15)$$

The responsibilities are computed using log-sum-exp for numerical stability. Positive q ensures that positive-weight components have positive updated means. An optional fixed weight floor $f \in [0, 1/K]$ replaces the unconstrained weight update in (2.15) by the exact optimizer of $\sum_a N_a \log \hat{\pi}_a$ on that floored simplex. If no floor is active, this reduces to N_a/n . The tested floor is given in Appendix C. Both exact versions maximize the variational M-step, giving objective monotonicity and boundedness within that stage. Proposition B.3 proves convergence of the objective within each fit; it does not prove global optimality or correct community recovery. Because the implementation reinitializes weights when it adds a component, we do not assert monotonicity between stages.

Algorithm 1: Spectral likelihood decoding with repeated replacement

1. Input K and the retained eigenpairs (U, Λ) ; form q by (2.6).
2. Obtain the global-mean growing fit by (2.11), (2.12), and the joint EM updates.
3. Evaluate the uniform-weight refit and all K global-gain replacements described in Section 2.4.
4. Accept the best proposal if its likelihood gain is at least $T_n/\sqrt{\log n}$ and return to step 3. Stop on rejection or after $\lceil (\log n)^2 \rceil$ accepted rounds.
5. Compute one E-step and one profile and weight M-step. Decode by (2.9), or continue exact EM before decoding.
6. Output labels, fitted profiles, weights, and the recorded objectives.

The archived experiments evaluate the original growing branch and a historical single replacement pass after partial peeling. The repeated procedure in Algorithm 1 additionally resets proposal weights uniformly, evaluates a uniform-weight refit, and iterates to the specified gain threshold. Its implementation checks are reported separately.

The numerical experiments use the exact floored-simplex weight update, finite stopping tolerances, and an iteration cap. Objective convergence in exact arithmetic does not replace checking the finite-run stopping rule or floating-point error. We report those numerical checks separately.

2.6 Why the representation can retain an LLR

Suppose temporarily that P and the other labels are known. For testing $z_i = a$ against $z_i = b$, define, for $j \neq i$,

$$w_j^{ab} = \log \frac{p_{aj}(1 - p_{bj})}{p_{bj}(1 - p_{aj})}, \quad c_i^{ab} = \sum_{j \neq i} \log \frac{1 - p_{aj}}{1 - p_{bj}}. \quad (2.16)$$

The oracle Bernoulli LLR is $L_i^{ab} = \sum_{j \neq i} A_{ij} w_j^{ab} + c_i^{ab}$. Extend w_i^{ab} by the same block formula. Because $A_{ii} = 0$, this extension does not change the oracle linear statistic. The contrast vector is block constant, hence belongs to $\text{col}(Z)$.

The fundamental identity is

$$\tilde{A} = A\Pi, \quad \Pi = UU^\top, \quad (\tilde{A}_i - A_i)w^{ab} = A_i(\Pi - \text{Id})w^{ab}. \quad (2.17)$$

Here Id denotes the identity matrix. If w^{ab} lies in the retained empirical space, the linear statistic is preserved exactly. At the population level, full rank gives $\text{col}(ZPZ^\top) = \text{col}(Z)$, so every contrast has the required span membership. The spectral decomposition need not itself optimize a likelihood: it retains a linear space containing the directions in which the likelihood acts.

The executable score additionally involves clipping, estimated profiles, mixture thresholds, and a retained reconstructed diagonal. These perturbations require statistical control. A matrix

reconstruction bound alone does not establish the oracle exponent; the local fitting theorem uses the stronger row-profile approximation, the clipping moment bound, and a weak-initialization condition. Theorem 3.6 supplies the fitting guarantee. Theorem 3.7 establishes its initialization condition for Algorithm 1 for every fixed K .

2.7 Computational cost

Forming the full positive reconstruction stores $O(n^2)$ numbers. Caching all candidate scores uses an n -by- n matrix and, by direct multiplication, $O(n^3)$ arithmetic. Each subsequent greedy scan costs $O(n^2)$, and one EM iteration costs $O(Kn^2)$. Algorithm 1 accepts $O(\sqrt{\log n})$ replacement rounds on the spectral event and has a deterministic $\lceil (\log n)^2 \rceil$ cap. Each round evaluates $K + 1$ fits and K candidate scans, so a prescribed polynomial EM iteration bound gives polynomial total cost. These costs are additional to computing the eigenpairs. The implementation is practical at the tested $n \leq 2000$ but does not yet justify a large-network speed claim. Candidate subsampling and blocked score evaluation are natural approximations, whose statistical and computational performance require separate validation.

3 Main recovery results

This section states the oracle benchmark, full-graph spectral approximation, and end-to-end guarantee for every fixed K . The model satisfies (2.2)–(2.3). The oracle calculations require distinct block rows; spectral conditioning supplies a uniformly separated K -dimensional signal.

3.1 Spectral information and likelihood contrasts

For communities $a \neq b$, define the block-constant Bernoulli contrast and the nonedge offset by

$$w_j^{ab} = \log \frac{p_{aj}(1 - p_{bj})}{p_{bj}(1 - p_{aj})}, \quad (3.1)$$

$$c_i^{ab} = \sum_{j \neq i} \log \frac{1 - p_{aj}}{1 - p_{bj}}. \quad (3.2)$$

The oracle incident-edge log-likelihood ratio is

$$L_i^{ab} = A_i w^{ab} + c_i^{ab}, \quad (3.3)$$

where A_i is row i of A . Its diagonal coordinate contributes zero.

Proposition 3.1 (Exact preservation of retained contrasts). *Let U contain any orthonormal collection of eigenvectors of the symmetric matrix A , let Λ contain their eigenvalues, and set $\Pi = UU^\top$ and $\tilde{A} = U\Lambda U^\top$. Then $\tilde{A} = A\Pi$ and, for every vector w ,*

$$\tilde{A}_i w - A_i w = A_i(\Pi - \text{Id})w. \quad (3.4)$$

In particular, $w \in \text{col}(U)$ implies exact preservation. Moreover, if P is nonsingular, then the rank- K comparison matrix $ZPZ^\top$ has column space $\text{col}(Z)$, and every w^{ab} belongs to this population signal space.

The proposition explains the likelihood interpretation without requiring the eigendecomposition itself to compute $\log P$. The eigendecomposition retains directions; the likelihood decoder assigns the appropriate contrast coefficients to those directions. It also states the precise limitation: the empirical retained space need not equal the population space. The actual no-self-loop mean is $\mathbb{E}[A | z] = ZPZ^\top - \text{diag}(P_{z_i z_i})$; the rank- K statement concerns $ZPZ^\top$, not the no-self-loop mean.

The first-order sparse Bernoulli contrast is $v_j^{ab} = \log(p_{aj}/p_{bj})$, with score

$$L_{i,\text{lin}}^{ab} = \sum_{j \neq i} A_{ij} v_j^{ab} - \sum_{j \neq i} (P_{a,z_j} - P_{b,z_j}). \quad (3.5)$$

The following result makes its use in a sparse working score precise.

Lemma 3.2 (Sparse Bernoulli likelihood expansion). *There is a constant C depending only on the fixed probability-ratio bounds such that, for all sufficiently large n , every i , and every $a \neq b$,*

$$|L_i^{ab} - L_{i,\text{lin}}^{ab}| \leq C\{P_{\max} \deg(i) + nP_{\max}^2\}. \quad (3.6)$$

For every fixed $H > 0$ there is a constant D_H such that $\Pr\{\deg(i) > D_H n P_{\max}\} \leq e^{-H n P_{\max}}$ uniformly in i . Consequently, the correction is $O(nP_{\max}^2) = o(nP_{\max})$ outside an event of probability at most $e^{-H n P_{\max}}$. It need not be $o(1)$. The condition $P_{\max} \rightarrow 0$ is used for this sparse-score approximation.

3.2 The oracle benchmark

We use Zhou–Li’s profile and Chernoff notation. For $\alpha \in [0, 1]$, define

$$D_\alpha(p_{a,-i} \| p_{b,-i}) = - \sum_{j \neq i} \log [p_{aj}^{1-\alpha} p_{bj}^\alpha + (1 - p_{aj})^{1-\alpha} (1 - p_{bj})^\alpha], \quad (3.7)$$

$$\alpha_{i,ab}^* \in \arg \max_{\alpha \in [0,1]} D_\alpha(p_{a,-i} \| p_{b,-i}), \quad D_{i,ab}^* = D_{\alpha_{i,ab}^*}(p_{a,-i} \| p_{b,-i}), \quad (3.8)$$

$$I_n = \min_{i, a \neq b} D_{i,ab}^*. \quad (3.9)$$

Here $p_{a,-i}$ omits the absent self-loop coordinate. The notation I_n is our shorthand for the minimum exact pairwise information; it is not a second definition of a rate.

Under (2.2)–(2.3), $I_n \asymp nP_{\max}$ and $I_n = \Omega(\log n)$. The normalized matrix may vary with n , so a limiting value of $I_n / \log n$ is not required. Zhou and Li [13] studied optimal error rates for SBM community detection; we use their information notation and leading exponent.

Lemma 3.3 (Leading oracle error exponent). *Consider the binary experiment in which $z_i \in \{a, b\}$ has equal prior probabilities and all other labels and P are known. Its Bayes risk $R_{i,ab}^{\text{or}}$ satisfies, uniformly in i ,*

$$R_{i,ab}^{\text{or}} = \exp\{-(1 + o(1))D_{i,ab}^*\}. \quad (3.10)$$

The same leading exponent holds for fixed prior probabilities bounded away from zero and one.

3.3 Transfer to estimated spectral scores

The theorem below accommodates scores estimated from the entire graph, so it does not assume independence of a node’s score and its incident edges. It makes the required rare-event control explicit.

Theorem 3.4 (Likelihood-ratio exponent transfer). *Let $\hat{\ell}_i(a)$ be estimated community scores, after one common alignment of their labels with the population labels, and classify by $\hat{z}_i \in \arg \max_a \hat{\ell}_i(a)$. Suppose there are $\tau_n = o(I_n)$ and $\delta_n \geq 0$ such that, for every i , an event G_i satisfies*

$$\Pr(G_i^c) \leq \delta_n, \quad (3.11)$$

$$\max_{a \neq b} |\hat{\ell}_i(a) - \hat{\ell}_i(b) - L_i^{ab}| \leq \tau_n \quad \text{on } G_i, \quad (3.12)$$

$$\liminf_{n \rightarrow \infty} \frac{-\log \delta_n}{I_n} \geq 1, \quad (3.13)$$

where $-\log 0 = +\infty$. Then the permutation-invariant error fraction obeys

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq \exp\{-(1 + o(1))I_n\}. \quad (3.14)$$

When $\liminf_{n \rightarrow \infty} I_n / \log n > 1$, this implies $\Pr\{\text{Mis}(\hat{z}, z) > 0\} \rightarrow 0$.

The condition on δ_n is essential. An error approximation that holds with probability $1 - o(1)$ can still fail much more frequently than the oracle makes an error, and therefore need not preserve its exponent. The theorem asserts an upper bound on the recovery error. In the binary oracle experiment, data processing and Lemma 3.3 also give the matching leading lower bound for any decoder using only the graph's spectral summary.

For the working scores actually fitted in Section 2, set $\hat{w}_j^{ab} = \log(\hat{p}_{aj}/\hat{p}_{bj})$ and $\hat{c}^{ab} = -\sum_j (\hat{p}_{aj} - \hat{p}_{bj}) + \log(\hat{\pi}_a/\hat{\pi}_b)$. Writing $q_i = \tilde{A}_i + (q_i - \tilde{A}_i)$ yields the exact decomposition

$$\begin{aligned} \hat{\ell}_i(a) - \hat{\ell}_i(b) - L_i^{ab} &= A_i(\Pi - \text{Id})w^{ab} \\ &\quad + \tilde{A}_i(\hat{w}^{ab} - w^{ab}) \\ &\quad + (q_i - \tilde{A}_i)\hat{w}^{ab} + (\hat{c}^{ab} - c_i^{ab}). \end{aligned} \quad (3.15)$$

Thus projection, estimated contrasts, clipping, and offsets must all be controlled at the scale in Theorem 3.4. The sum includes the reconstructed diagonal used by the implemented working score; its discrepancy with the oracle's excluded diagonal is included in the displayed decomposition. Neither average reconstruction accuracy nor objective convergence removes these terms.

3.4 A local recovery theorem for the empirical spectral fit

The following approximation controls the empirical reconstruction computed from the whole graph. Set $\Pi_* = ZN^{-1}Z^\top$ and

$$q^0 = \max\{A\Pi_*, \varepsilon_n\}, \quad q = \max\{U\Lambda U^\top, \varepsilon_n\},$$

with the observable floor from (2.6). The comparison q^0 uses the true memberships only in the analysis.

Lemma 3.5 (Full-graph spectral approximation). *Under the model above with $0 < c_0 < c_r/2$, for every fixed $H > 0$ there are constants $c_H, C_H > 0$ and a deterministic sequence $a_n \rightarrow 0$ such that, with probability at least $1 - O(e^{-HnP_{\max}})$,*

$$\max_i \|q_i - q_i^0\|_1 \leq a_n n P_{\max}, \quad c_H P_{\max} \leq q_{ij} \leq C_H P_{\max}, \quad (3.16)$$

$$\frac{1}{n^2 P_{\max}} \sum_i \|q_i - p_{z_i, *}\|_1 \leq a_n. \quad (3.17)$$

The reconstruction uses the empirical eigenpairs of the whole graph. No row-independent replacement or extra leave-one-out computation is required by the algorithm.

The entrywise eigenspace theorem of Abbe et al. [2], the bounded-entry norm tail of Bandeira and van Handel [4], and a direct Bennett row bound supply these inputs. The appendix verifies the assumptions at failure probability $e^{-HnP_{\max}}$. This confidence scale matters when expected degrees grow faster than $\log n$.

Theorem 3.6 (Local empirical spectral EM attains the leading exponent). *Use the model and admissible observable floor of Lemma 3.5. Let the initial responsibilities $R^{(0)} = (R_{ia}^{(0)})$ have nonnegative rows summing to one. They may depend on the entire graph. Suppose that, after a common label alignment, for deterministic $\gamma_n \rightarrow 0$ and $\delta_n \geq 0$,*

$$\Pr \left\{ \frac{1}{n} \sum_{i,a} |R_{ia}^{(0)} - Z_{ia}| > \gamma_n \right\} \leq \delta_n, \quad \liminf \frac{-\log \delta_n}{I_n} \geq 1. \quad (3.18)$$

Perform at least one responsibility-weighted profile M-step on q , followed by the score classification in (2.9). Then

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}. \quad (3.19)$$

The same bound holds after any further exact EM updates and any finite, possibly data-dependent stopping time after the first M-step. The weights are updated either unconstrained or by the exact simplex optimizer with a fixed floor $f \in [0, 1/K]$. If a component has zero responsibility mass, retain its previous profile; on the weak-start event this case does not occur for large n . When $\liminf_{n \rightarrow \infty} I_n / \log n > 1$, the output exactly recovers all labels with probability tending to one.

The local theorem controls dependent reconstructed profiles and freely fitted component means. The next theorem establishes its initialization condition through the repeated-replacement certificate.

3.5 End-to-end recovery for every fixed order

Theorem 3.7 (Spectral likelihood decoding for arbitrary fixed K). *For every fixed integer $K \geq 2$, under (2.2)–(2.3) and $0 < c_0 < c_r/2$, Algorithm 1 satisfies*

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}.$$

If $\liminf I_n / \log n > 1$, it recovers all labels with probability tending to one. The growing initialization, replacements, and refinement access only (U, Λ) . Every proposal begins with uniform weights and may use any prescribed finite number of exact EM updates. No assumed weak initializer or global mixture optimum is required. A polynomial EM iteration cap gives polynomial runtime.

The guarantee covers the repeated replacement rule in Section 2.4. It follows from the deterministic certificate in Lemma A.3, then the mandatory final M-step and Theorem 3.6. The original growing branch is retained as its initializer. A separate appendix result records the guarantee for that branch alone when $K = 2$.

4 Experiments

The archived 240-graph comparisons evaluate the original growing branch of Section 2.3 and the separately specified controls. They do not evaluate the complete repeated-replacement Algorithm 1 on that full collection. The repeated replacement is a proved algorithmic extension; its separate implementation verification is documented in the repository.

We evaluate whether likelihood-based decoding improves on a Euclidean decoder of the same retained eigenpairs, and whether global likelihood gains prevent redundant initialization. The experiments use a spectral profile working likelihood. They do not evaluate an exact leave-one-out graph likelihood or estimate an asymptotic error exponent from finite samples.

4.1 Models and protocol

For each model, let $C^{(0)}$ be a fixed symmetric connectivity matrix and let π be the vector of community proportions. We generate an undirected SBM without self-loops, with edge probabilities

$$P_{ab} = \frac{\log}{n} n C_{ab}, \quad C = \frac{\eta}{D_{\min}(C^{(0)}, \pi)} C^{(0)}, \quad (4.1)$$

where $\eta \in \{0.8, 1.4\}$ and

$$D_{\min}(C, \pi) = \min_{a \neq b} \max_{0 \leq t \leq 1} \sum_{c=1}^K \pi_c \left[(1-t) C_{ac} + t C_{bc} - C_{ac}^{1-t} C_{bc}^t \right]. \quad (4.2)$$

Thus the limiting minimum Chernoff–Hellinger coefficient equals η . The normalization uses the limiting proportions π ; actual group sizes are $\lfloor n\pi_a \rfloor$, with the remaining vertices assigned to the last group. Vertex labels are then randomly permuted. Each model is tested at $n \in \{1000, 2000\}$. The six models are

$$\begin{aligned} C_{\text{sym2}}^{(0)} &= \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}, & \pi_{\text{sym2}} &= (1/2, 1/2), \\ C_{\text{equal3}}^{(0)} &= \begin{pmatrix} 10 & 1 & 2 \\ 1 & 11 & 1 \\ 2 & 1 & 10 \end{pmatrix}, & \pi_{\text{equal3}} &= (1/3, 1/3, 1/3), \\ C_{\text{hetero3}}^{(0)} &= \begin{pmatrix} 6 & 0.5 & 0.5 \\ 0.5 & 12 & 0.5 \\ 0.5 & 0.5 & 24 \end{pmatrix}, & \pi_{\text{hetero3}} &= (1/3, 1/3, 1/3), \\ C_{\text{unequal3}}^{(0)} &= \begin{pmatrix} 10 & 1 & 1 \\ 1 & 10 & 1 \\ 1 & 1 & 10 \end{pmatrix}, & \pi_{\text{unequal3}} &= (0.5, 0.3, 0.2), \\ C_{\text{hier4}}^{(0)} &= \begin{pmatrix} 12 & 2 & 0.5 & 0.5 \\ 2 & 12 & 0.5 & 0.5 \\ 0.5 & 0.5 & 12 & 2 \\ 0.5 & 0.5 & 2 & 12 \end{pmatrix}, & \pi_{\text{hier4}} &= (1/4, 1/4, 1/4, 1/4), \\ C_{\text{dis3}}^{(0)} &= \begin{pmatrix} 0.5 & 8 & 8 \\ 8 & 0.5 & 8 \\ 8 & 8 & 0.5 \end{pmatrix}, & \pi_{\text{dis3}} &= (1/3, 1/3, 1/3). \end{aligned}$$

These include unequal group sizes, heterogeneous expected degrees, hierarchical connections, and negative signal eigenvalues.

The design batch contains five graph seeds, 20000, ..., 20004, for every model, size, and signal coefficient, giving $6 \times 2 \times 2 \times 5 = 120$ graphs. A second batch uses the same models and hyperparameters with fresh seeds 30000, ..., 30004. The algorithms and second-batch seed range were fixed before fitting the second batch. The second batch therefore validates initialization choices made using the first batch, rather than introducing new models. All methods are evaluated on identical eigenpairs within a graph.

We retain the K eigenpairs of the raw adjacency matrix with largest absolute eigenvalues. The implementation requests $K + 1$ eigenpairs using the symmetric ARPACK solver with tolerance 10^{-8} and discards the last one after sorting by absolute value. After this preprocessing, the fitting routines receive only (U, Λ) ; adjacency entries, degrees, true labels, and true block probabilities are unavailable to fitting and model selection.

4.2 Implementations and comparisons

The archived implementation forms positive spectral profiles

$$q_{ij} = \max \left\{ (U \Lambda U^\top)_{ij}, 10^{-4} \frac{\log n}{n} \right\}, \quad \ell_i(\hat{p}) = \sum_{j=1}^n q_{ij} \log \hat{p}_j - \sum_{j=1}^n \hat{p}_j. \quad (4.3)$$

The reconstructed diagonal is retained, even though the generated graphs have zero diagonal. Thus this is a declared positive-profile working score, not the original Bernoulli graph likelihood. Component means are unrestricted n -dimensional profile means. Every likelihood arm uses the same soft EM updates and observed working objective

$$\mathcal{L}(\hat{\pi}, \hat{p}) = \sum_{i=1}^n \log \left[\sum_{a=1}^K \hat{\pi}_a \exp\{\ell_i(\hat{p}_a)\} \right]. \quad (4.4)$$

EM initializes equal component weights except when refitting a replacement proposal, where the current weights are retained. The mean update is the responsibility-weighted average of q_i . The weight update exactly maximizes the EM auxiliary objective on $\{\hat{\pi}_a \geq 10^{-8}, \sum_a \hat{\pi}_a = 1\}$. No additional floor on means is needed because the observations are positive. Final labels maximize $\log \hat{\pi}_a + \ell_i(\hat{p}_a)$.

The original deterministic growing branch of Section 2.3 is recorded as `growing_global_gain_EM`. It starts from the global profile mean, adds $K - 1$ node candidates by their all-node marginal gains, and runs joint EM after each addition. It requires no top-subset size and removes no nodes.

The frozen-profile control, recorded as `global_gain_no_update`, starts with the historical top-subset rule and then chooses all remaining node seeds by all-node gains, keeping the profiles frozen until one final EM fit. Its exact historical subset sizes are recorded in Appendix C. It is a different algorithm despite its identical recovery errors in these experiments.

We compare it with the following alternatives. *Top-subset peeling* uses the historical first-seed rule and repeatedly removes the selected candidate and its best-scoring subset, using the archived sizes specified in the appendix. After K selections, all n observations, including unselected nodes, enter EM. *Peeling plus replacement* fits this initialization, deletes each component temporarily, proposes the globally most useful node against the remaining $K - 1$ means, and refits all nodes. All K proposals use the same original fitted endpoint; the best proposal is accepted only if its final $\mathcal{L}$ exceeds the current best by more than 10^{-5} . This is one replacement pass, not repeated optimization.

Table 1: Average mislabeled fraction in percent; exact recovery counts are out of 120 graphs per batch. All profile arms use the same EM score.

Method	Design batch		Independent batch	
	Error (%)	Exact	Error (%)	Exact
Top-subset peeling + EM	0.3917	86	0.3725	77
Peeling + replacement	0.0967	86	0.0925	77
Frozen global gain + EM	0.0967	86	0.0925	77
Growing global gain	0.0967	86	0.0925	77
Growing sampled gain + EM	0.3900	86	0.0925	77
Residual++, one start + EM	4.0129	80	3.1675	72
Greedy residual++, one start + EM	1.1192	85	2.3438	74
Greedy residual++, ten starts + EM	0.0967	86	0.0925	77
Euclidean++, one start + EM	3.9033	78	6.4204	71
X k -means, ten starts	0.3454	73	0.2933	66

The API option `method='global_gain'` invokes the primary growing algorithm; it does not invoke the frozen-profile control. *Residual++* samples the first node uniformly and later nodes proportionally to

$$d_i = \min_{a \in S} \sum_j [q_{ij} \log(q_{ij}/\hat{p}_{aj}) - q_{ij} + \hat{p}_{aj}]. \quad (4.5)$$

This deviance is not squared again. Its greedy variant samples $2 + \lfloor \log K \rfloor$ candidate trials per addition and chooses the trial that minimizes total residual deviance. We report both one start and ten starts; the ten-start fit is selected solely by final $\mathcal{L}$. An additional control selects one set of Euclidean k -means++ seeds in $X = \sqrt{n}U\Lambda$ and then runs the same profile EM. The main Euclidean baseline runs k -means in X with ten starts, at most 300 Lloyd iterations per start, and tolerance 10^{-5} .

All profile EM fits use at most 300 updates and tolerance 10^{-6} , with label, parameter-change and objective-change checks. The exact archived stopping rule is recorded in Appendix C. Randomized seeding uses fixed NumPy seed sequences indexed by graph seed, method, and restart. Reported error rates minimize label disagreement over all permutations of the K labels. Averages weight graphs equally; “exact” counts graphs with no mislabeled node. True labels are accessed only after all fitting and likelihood-based selections finish.

4.3 Recovery results

Table 1 gives the paired results in the design and independent batches. Global-gain initialization, peeling plus replacement, growing global gain, and ten-start greedy residual++ have the same per-graph error counts in both batches. Across all 240 graphs, the main deterministic method has average error 0.0946%, compared with 0.3194% for the ten-start Euclidean baseline, and exactly recovers 163 graphs compared with 139. These are finite-sample recovery comparisons with unequal optimization budgets, rather than equal-runtime comparisons.

Table 2 separates model families. The largest improvement over Euclidean decoding occurs with heterogeneous diagonal connectivities: the average error falls from 1.4175% to 0.1625%. The likelihood criterion also improves the unequal-size model. It is not uniformly better: the symmetric two-community baseline has one fewer mislabeled node in aggregate, and the hierarchical model has one additional exactly recovered graph under k -means. The examples therefore support likelihood adaptation when block profiles differ substantially, without claiming dominance on every graph.

Table 2: Both batches combined, 40 graphs per model. The main columns use the original growing global-gain branch. The frozen-profile control has identical recorded error counts.

Model	Growing global gain + EM		X k -means	
	Error (%)	Exact	Error (%)	Exact
Symmetric two-community	0.0150	33	0.0125	34
Equal-size three-community	0.0175	32	0.0188	31
Heterogeneous three-community	0.1625	18	1.4175	1
Unequal-size three-community	0.0338	33	0.1213	25
Hierarchical four-community	0.0275	28	0.0288	29
Disassortative three-community	0.3113	19	0.3175	19

The stronger signal coefficient $\eta = 1.4$ gives 117 exactly recovered graphs out of 120 for global gain, versus 99 for k -means. At $\eta = 0.8$, the corresponding counts are 46 and 40. This behavior is consistent with the importance of the logarithmic-density recovery threshold, but the experiment does not establish that threshold or the leading error exponent.

4.4 A failure of subset scores and its repair

One design graph with $n = 2000$, $K = 3$, and $\eta = 0.8$ illustrates the initialization mechanism. Its normalized connectivity matrix is approximately

$$C = \begin{pmatrix} 7.8541 & 0.7854 & 1.5708 \\ 0.7854 & 8.6395 & 0.7854 \\ 1.5708 & 0.7854 & 7.8541 \end{pmatrix}, \quad (4.6)$$

with group sizes (666, 666, 668). Communities 1 and 3 are the closest pair: their empirical X -centroid distance is 40.22, compared with 47.36 and 47.82 for the other pairs. They are nevertheless well separated in the retained spectrum. Using their true profile means as a diagnostic classifier assigns every node correctly on its first assignment.

Top-subset peeling selects nodes 858, 610, 435 (zero-based IDs), whose true communities are 2, 3, 2. The profile of node 610 lies near the mixture of communities 1 and 3. Its profile divergence to their merged mean is 0.3826, compared with 5.2433 to its own community’s mean. Its selected subset has 150 nodes from community 1 and 196 from community 3. Thus a candidate can win a subset likelihood comparison by representing a mixture of communities. The resulting fit merges communities 1 and 3 and splits community 2, making 709 errors (35.45%). Tightening the tolerance to 10^{-10} and permitting 3000 updates does not solve this structural problem: the fit stops after 517 updates with 800 errors.

The frozen global-gain control keeps the first two seeds but changes the third to node 220, which contributes to the uncovered region; after EM, only one node is mislabeled. A one-component replacement at the original fitted endpoint selects the same node and likewise reduces errors to one, while increasing the working log likelihood by approximately 7148.48. No true labels enter either choice. The independent batch contains a second analogous peeling failure with 675 errors, reduced to three by replacement. Relative to peeling, each effective deterministic variant improves exactly one graph in each batch, leaves the other 119 error counts unchanged, and worsens none. The number of exact recoveries does not change because the repaired graphs retain one and three errors, respectively.

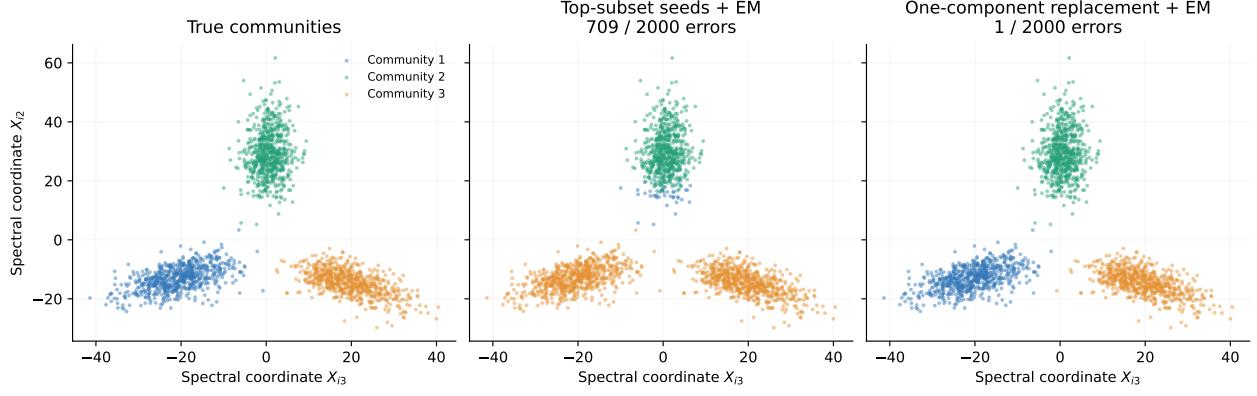


Figure 1: The saved design failure, displayed in two coordinates of $X = \sqrt{n}U\Lambda$. Left: true communities, used only for evaluation. Middle: top-subset peeling followed by EM merges the two lower communities and splits the upper community. Right: an accepted one-component replacement followed by EM separates the groups and leaves one mislabeled node. Predicted labels are aligned to truth for display only.

4.5 Numerical verification and computational scope

All returned profile fits satisfy the declared stopping rule in the 240-graph comparison. The 120 original peeling errors are reproduced exactly. The archive contains 6240 EM traces, including replacement proposals, growing stages, and restart fits, with 27867 recorded trace points. The smallest observed objective increment is -1.17×10^{-10} , within floating-point rounding; no increment below -10^{-7} occurs. All 240 selected ten-start solutions maximize the same final working likelihood within their restart set. Replacement accepts a likelihood improvement on 22 design graphs and 25 independent graphs, and never lowers the final objective; most of these small improvements do not change the classification error. Independent scalar checks reproduce deviance and global-gain calculations with maximum absolute errors 7.11×10^{-15} and 1.78×10^{-15} , respectively.

The present implementation explicitly reconstructs the $n \times n$ profiles and caches all singleton-candidate scores. Reconstruction costs $O(n^2K)$, score precomputation costs $O(n^3)$, and storage costs $O(n^2)$. An EM update costs $O(n^2K)$. A replacement pass tries K additional fits, whereas ten-start residual seeding uses ten fits. The runs use one numerical library thread per worker and four worker processes. Python, NumPy, SciPy, and scikit-learn versions are 3.11.7, 2.1.3, 1.16.2, and 1.7.1. Stored timers cover all candidate arms and restarts together and exclude graph generation and eigenpair computation, so they do not support an arm-specific speed claim. The data sizes $n \leq 2000$ establish feasibility for this implementation; candidate-pool approximations and blockwise computations require separate large-graph validation.

The evidence supports improved decoding of a fixed spectrum and a repair of redundant initialization under the declared working likelihood. It does not by itself prove that the practical EM endpoint has the oracle error exponent, that clipping preserves exact graph likelihood ratios, or that the growing initializer meets the high-probability weak-start condition in Theorem 3.6. Oracle-LRT comparisons, a denser range of n for error-exponent estimation, and a full evaluation of Algorithm 1 remain outside the archived numerical study.

4.6 Fresh checks of the fixed- K certified procedure

The new implementation was checked on 24 SBM graphs with $K \in \{3, 4, 6\}$, $n \in \{256, 512\}$, two graph seeds for each configuration, and mean expected-degree targets $4 \log n$ or $2\sqrt{n}$. Community sizes are balanced up to rounding, but the symmetric block matrices have unequal row sums, hence unequal expected degrees. Fitting receives only the retained eigenpairs. Truth is used afterwards to report errors.

K	Mean degree target	Graphs	Exact graphs	Errors/nodes
3	$4 \log n$	4	4	0/1536
3	$2\sqrt{n}$	4	4	0/1536
4	$4 \log n$	4	3	1/1536
4	$2\sqrt{n}$	4	4	0/1536
6	$4 \log n$	4	0	6/1536
6	$2\sqrt{n}$	4	4	0/1536

All 24 runs stopped by rejecting every proposal at the specified threshold, and every proposal started from uniform weights. There were 7 mislabeled nodes among 9216 nodes, with 19 exact graph recoveries. The smallest recorded within-fit objective increment was -1.46×10^{-11} , consistent with floating-point rounding. An additional diagnostic started the replacement procedure from an injected three-component collapsed fit: it accepted two consecutive replacements and then reached its stopping certificate with zero label errors. Checks also covered a zero spectrum, preservation of the historical floor option, and the public deterministic API. These finite samples verify the implemented procedure and its stopping logic; they do not estimate its asymptotic exponent. The script and machine-readable results are archived with the implementation.

5 Conclusion

Spectral compression can preserve the likelihood information needed for nearly optimal community recovery. In a full-rank SBM, the membership subspace contains the block-constant likelihood contrasts. First-order analysis of the empirical adjacency eigenpairs makes this relation quantitative when the minimum expected degree is $\Omega(\log n)$, and an admissible spectral-scale positivity floor preserves the leading Chernoff exponent. Expected degrees can vary across communities, and the sparse block probabilities need not have a fixed normalized shape.

For every fixed $K \geq 2$, Algorithm 1 combines global-mean growing initialization with deterministic repeated component replacement and a final EM update. The stopping certificate establishes weakly correct responsibilities, and the local refinement theorem gives

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}.$$

Initialization and likelihood selection are part of the proof. The guarantee matches the leading oracle exponent and yields exact recovery above the corresponding logarithmic information threshold. The original single-pass growing procedure is a separate algorithm; its two-community guarantee is recorded in the appendix.

The paired 240-graph experiments evaluate the original growing and historical parameter-replacement procedures. They support the usefulness of all-node likelihood gains and likelihood-accepted repairs in resolving severe initialization failures. The new repeated-replacement algorithm

has additional implementation checks, but a broad accuracy benchmark remains to be carried out. Reducing the cost of full profile reconstruction and establishing a general fixed- K theorem for growing initialization alone are further questions.

A Proofs of spectral preservation and statistical results

A.1 Exact preservation and the population contrast space

Proof of Proposition 3.1. Since $AU = U\Lambda$, multiplication by $U^\top$ gives $A\Pi = AUU^\top = U\Lambda U^\top = \tilde{A}$. Therefore $\tilde{A}_i w - A_i w = A_i(\Pi - \text{Id})w$. If $w \in \text{col}(U)$, then $\Pi w = w$ and the difference vanishes.

Let $N = Z^\top Z = \text{diag}(n_1, \dots, n_K)$, which is nonsingular since all communities are nonempty. The columns of $ZN^{-1/2}$ are orthonormal, and

$$ZPZ^\top = (ZN^{-1/2})(N^{1/2}PN^{1/2})(ZN^{-1/2})^\top.$$

Nonsingularity of P makes the central matrix nonsingular. Hence $ZPZ^\top$ has rank K and column space $\text{col}(Z)$. For each contrast, define $\xi_c^{ab} = \log\{P_{ac}(1 - P_{bc})/[P_{bc}(1 - P_{ac})]\}$. Then $w^{ab} = Z\xi^{ab}$, which proves membership in that space. Finally, subtracting $\text{diag}(P_{z_i z_i})$ from $ZPZ^\top$ gives the actual no-self-loop mean, so no rank assertion about that corrected matrix is needed. $\square$

Full rank cannot be replaced by distinct block rows without a further contrast-span condition. For example, let $P = ss^\top$ for a positive, nonconstant vector s . Its signal space is spanned by Zs , whereas the sparse log-probability contrast between communities a, b is $\log(s_a/s_b)\mathbf{1}$. When $s_a \neq s_b$, this constant vector does not belong to $\text{span}(Zs)$. A rank-one signal projection does not preserve that contrast by the preceding identity. This example only concerns projection onto the population rank-one space; it is not an impossibility claim for every statistic computed from one empirical eigenvector.

A.2 The sparse likelihood correction

Proof of Lemma 3.2. Write $p_{aj} = P_{a,z_j}$ and $p_{bj} = P_{b,z_j}$ for $j \neq i$. The exact binary Bernoulli log-likelihood ratio is

$$L_i^{ab} = \sum_{j \neq i} A_{ij} \log \frac{p_{aj}}{p_{bj}} + \sum_{j \neq i} A_{ij} \log \frac{1 - p_{bj}}{1 - p_{aj}} + \sum_{j \neq i} \log \frac{1 - p_{aj}}{1 - p_{bj}}.$$

For $0 \leq x \leq 1/2$, the power series for $\log(1 - x)$ implies $|\log(1 - x) + x| \leq Cx^2$. Since $p_{aj}, p_{bj} \leq P_{\max} \rightarrow 0$,

$$\left| \log \frac{1 - p_{bj}}{1 - p_{aj}} \right| \leq CP_{\max}, \quad \left| \log \frac{1 - p_{aj}}{1 - p_{bj}} + (p_{aj} - p_{bj}) \right| \leq CP_{\max}^2.$$

The first term is unchanged in the sparse score, and the final term is replaced by its first-order nonedge offset. Summing proves (3.6).

Under any true community, $\deg(i)$ is a sum of independent Bernoulli variables with mean at most $nP_{\max}$. For $D > 1$, exponential Markov inequality with $t = \log D$ gives

$$\Pr\{\deg(i) > DnP_{\max}\} \leq \exp\{-nP_{\max}(D \log D - D + 1)\}.$$

The bracket tends to infinity with D . Thus, for every fixed $H > 0$, the degree bound holds uniformly in i outside probability $e^{-HnP_{\max}}$; a union bound has the same form after increasing D , because $nP_{\max} \geq c \log n$. On that event the likelihood correction is $O_H(nP_{\max}^2) = o(nP_{\max})$. This error need not tend to zero in absolute value: its relevant scale is the information $I_n \asymp nP_{\max}$ established below. $\square$

Remark A.1 (Scope of the sparse working score). The density lower bound permits expected degrees larger than $\log n$. The assumption $P_{\max} \rightarrow 0$ is used separately to make the first-order Bernoulli correction negligible relative to I_n . No result here identifies this sparse score with the exact Bernoulli likelihood when the block probabilities stay bounded away from zero.

A.3 Chernoff information and the oracle exponent

Proof of Lemma 3.3. Let f_a, f_b denote the product Bernoulli mass functions of the incident-edge row under the two hypotheses. Define

$$\begin{aligned} D_\alpha(p_{a,-i} \| p_{b,-i}) &= -\log \sum_x f_a(x)^{1-\alpha} f_b(x)^\alpha \\ &= -\sum_{j \neq i} \log \{p_{aj}^{1-\alpha} p_{bj}^\alpha + (1-p_{aj})^{1-\alpha} (1-p_{bj})^\alpha\}, \\ D_{i,ab}^* &= \max_{0 \leq \alpha \leq 1} D_\alpha(p_{a,-i} \| p_{b,-i}). \end{aligned}$$

Uniformly for $\alpha \in [0, 1]$ and $u, v \in [P_{\min}, P_{\max}]$,

$$u^{1-\alpha} v^\alpha + (1-u)^{1-\alpha} (1-v)^\alpha = 1 - \{(1-\alpha)u + \alpha v - u^{1-\alpha} v^\alpha\} + O(P_{\max}^2).$$

For example, the nonedge product expansion follows by writing it as $\exp\{(1-\alpha)\log(1-u) + \alpha\log(1-v)\}$. Expanding $-\log(1-x)$ and summing gives

$$D_{i,ab}^* = n \max_{0 \leq \alpha \leq 1} \sum_c \pi_{c,n} \{(1-\alpha)P_{ac} + \alpha P_{bc} - P_{ac}^{1-\alpha} P_{bc}^\alpha\} + O(nP_{\max}^2 + P_{\max}), \quad (\text{A.1})$$

where $\pi_{c,n} = n_c/n$. The $O(P_{\max})$ term accounts for the excluded node. A fixed limiting normalized matrix is unnecessary. Indeed, $\sigma_{\min}(N^{1/2}PN^{1/2}) \geq cnP_{\max}$ implies $\sigma_{\min}(P) \geq cP_{\max}$, since $\|N^{1/2}\|_{\text{op}}^2 \leq n$. Consequently, for every $a \neq b$,

$$\sum_c \pi_{c,n} (\sqrt{P_{ac}} - \sqrt{P_{bc}})^2 \geq \frac{\pi_{\min}}{4P_{\max}} \|P_{a*} - P_{b*}\|_2^2 \geq c' P_{\max}.$$

Here P_{a*} denotes the K -coordinate block row in this display. Taking $\alpha = 1/2$ in (A.1) proves a uniform lower bound $D_{i,ab}^* \geq c'nP_{\max}$ for large n . The upper bound $D_{i,ab}^* \leq CnP_{\max}$ follows from the same expansion. Thus $D_{i,ab}^* \asymp nP_{\max}$ and $I_n \asymp nP_{\max}$, uniformly over all pairs and excluded nodes.

The equal-prior Bayes risk is $R_{i,ab}^{\text{or}} = \frac{1}{2} \sum_x \min\{f_a(x), f_b(x)\}$. Since $\min\{u, v\} \leq u^{1-\alpha} v^\alpha$,

$$R_{i,ab}^{\text{or}} \leq \frac{1}{2} e^{-D_{i,ab}^*}. \quad (\text{A.2})$$

A matching leading lower bound follows by exponential tilting. Let α_* maximize the Chernoff information and set

$$f_*(x) = e^{D_{i,ab}^*} f_a(x)^{1-\alpha_*} f_b(x)^{\alpha_*}, \quad L(x) = \log \frac{f_a(x)}{f_b(x)}.$$

The two positive product distributions differ and the endpoint derivatives have opposite signs. Hence $\alpha_* \in (0, 1)$, and differentiating the normalization gives $\mathbb{E}_* L = 0$. The tilted incident edges remain independent, with probabilities

$$p_{*,j} = \frac{p_{aj}^{1-\alpha_*} p_{bj}^{\alpha_*}}{p_{aj}^{1-\alpha_*} p_{bj}^{\alpha_*} + (1-p_{aj})^{1-\alpha_*} (1-p_{bj})^{\alpha_*}} \leq CP_{\max}.$$

The edge coefficient in L is uniformly bounded because $P_{\min} \geq cP_{\max}$ and $P_{\max} \rightarrow 0$. Thus $\text{Var}_*(L) \leq CnP_{\max}$. Take $h_n = (nP_{\max})^{2/3}$. Chebyshev's inequality gives $\Pr_*(|L| \leq h_n) = 1 - o(1)$, and the change of measure yields

$$\begin{aligned} R_{i,ab}^{\text{or}} &= \frac{1}{2} e^{-D_{i,ab}^*} \mathbb{E}_* \min\{e^{\alpha_* L}, e^{-(1-\alpha_*)L}\} \\ &\geq \frac{1}{2} e^{-D_{i,ab}^* - h_n} \Pr_*(|L| \leq h_n). \end{aligned}$$

Together with (A.2), this proves $\log R_{i,ab}^{\text{or}} = -D_{i,ab}^* + o(nP_{\max})$ and hence (3.10). Fixed prior probabilities bounded away from zero and one change this risk by at most fixed factors, preserving the leading exponent. $\square$

A.4 Transfer of the leading exponent

Proof of Theorem 3.4. Suppose the true label of node i is a . If G_i occurs and the classifier prefers $b \neq a$, then $\hat{\ell}_i(a) - \hat{\ell}_i(b) \leq 0$, and consequently $L_i^{ab} \leq \tau_n$. For every $\alpha \in [0, 1]$,

$$\Pr\{L_i^{ab} \leq \tau_n\} \leq e^{\alpha\tau_n} \mathbb{E}_a e^{-\alpha L_i^{ab}} = \exp\{\alpha\tau_n - D_\alpha(p_{a,-i} \| p_{b,-i})\}.$$

Selecting a Chernoff maximizer and using $\alpha \leq 1$ gives

$$\Pr\{\hat{z}_i \neq z_i\} \leq \delta_n + \sum_{b \neq a} \exp\{\tau_n - D_{i,ab}^*\} \leq \delta_n + (K-1) \exp\{-I_n + o(I_n)\}.$$

Because $\tau_n = o(I_n)$, (3.13) makes the first term no larger in leading exponential order. Averaging over nodes and using that the permutation-invariant error is no larger than the error under any one common alignment proves (3.14).

The expected number of errors under that alignment is at most $n \exp\{-(1+o(1))I_n\}$. If $\liminf_{n \rightarrow \infty} I_n / \log n > 1$, Markov's inequality gives vanishing probability of even one error, which proves the exact recovery assertion. No conclusion when $I_n / \log n \rightarrow 1$ follows from this argument. $\square$

The matching binary lower bound mentioned after the theorem has a direct decision-theoretic justification. Conditional on all other labels, the rest of the graph is independent of the incident-edge row and has the same distribution under a and b . Revealing the other labels and true parameters cannot increase the Bayes risk. A function of the graph's spectral summary is also a function of the full graph; its risk in that binary experiment is therefore at least $R_{i,ab}^{\text{or}}$. This argument does not, by itself, assert a global minimax lower bound over an unspecified parameter space.

A.5 Full-graph reconstruction and local empirical EM

Proof of Lemma 3.5. The true no-self-loop mean $\mathbb{E}[A \mid z]$ leaves $\text{col}(Z)$ invariant. In the orthonormal membership basis $ZN^{-1/2}$ its restriction is $N^{1/2}PN^{1/2} - \text{diag}(P_{11}, \dots, P_{KK})$. The conditioning

assumption implies that its K signal eigenvalues have absolute values between $cnP_{\max}$ and $CnP_{\max}$ for large n . On the orthogonal complement, the eigenvalues are $-P_{aa}$, with absolute values at most $P_{\max}$. Let U_*, Λ_* contain the signal eigenpairs. Then $U_*U_*^\top = \Pi_*$ and $\|U_*\|_{2 \rightarrow \infty} = O(n^{-1/2})$.

Here $\|M\|_{2 \rightarrow \infty} = \max_i \|M_i\|_2$. Fix any $H > 0$. The bounded-entry spectral norm inequality of Bandeira and van Handel [4, Corollary 3.12 and Remark 3.13] gives, for $t \geq 0$ and absolute constants,

$$\Pr\{\|A - \mathbb{E}[A \mid z]\|_{\text{op}} > C\sqrt{nP_{\max}} + t\} \leq ne^{-t^2/C}.$$

Taking $t = C_H\sqrt{nP_{\max}}$ absorbs the factor n because $nP_{\max} \geq c \log n$. Thus the noise norm is at most $C_H\sqrt{nP_{\max}}$ outside $O(e^{-HnP_{\max}})$. In the notation of Abbe et al. [2, Theorem 2.1], the signal separation is comparable to $nP_{\max}$, the eigenvalue condition number is bounded, and we take $\gamma = C_H/\sqrt{nP_{\max}}$. Their incoherence assumption holds since $\|\mathbb{E}[A \mid z]\|_{2 \rightarrow \infty} = O(nP_{\max}/\sqrt{n})$ and $nP_{\max} \leq n$. Row/column independence follows from the independent upper-triangular edges.

We verify their row concentration at the same exponential confidence. For a fixed matrix W with at most K columns, put $m = \|W\|_{2 \rightarrow \infty}$ and $x = \|W\|_F/(\sqrt{nm}) \in (0, 1]$; $m = 0$ is trivial. Every scalar column sum of $(A - \mathbb{E}[A \mid z])_{i*}W$ has summands bounded by m and variance at most $nP_{\max}m^2x^2$. Bennett's inequality at threshold $BnP_{\max}m/(1 \vee \log(1/x))$ gives an exponent at least

$$nP_{\max}x^2h\left(\frac{B}{x^2(1 \vee \log(1/x))}\right) \geq \frac{B}{2}nP_{\max}, \quad h(u) = (1+u)\log(1+u) - u,$$

for all $0 < x \leq 1$ and all sufficiently large fixed B . For $x \leq e^{-1}$ this follows from

$$\log\{B/[x^2 \log(1/x)]\} \geq \log(1/x), \quad h(u) \geq u \log(u)/2$$

for sufficiently large u . For $x \geq e^{-1}$ it follows by monotonicity of the Bennett exponent in its variance bound. Choosing B sufficiently large and applying a union bound over the fixed number of columns proves their A4 with per-row failure at most $e^{-HnP_{\max}}/n$ and

$$\varphi_H(x) = \frac{C_H}{1 \vee \log(1/x)} \quad (x > 0), \quad \varphi_H(0) = 0.$$

The function is continuous and nondecreasing, and $\varphi_H(x)/x$ is nonincreasing. Both γ and $\varphi_H(\gamma)$ tend to zero, so their smallness assumption holds for large n . We may take both failure parameters in their theorem to be $O(e^{-HnP_{\max}})$, after adjusting the constants.

Apply that theorem separately to the positive and negative signal groups and concatenate the polar orthogonal alignments. Repeated eigenvalues within either group cause no difficulty. The theorem allows a non-low-rank mean, so the diagonal correction has been retained. The noise norm event identifies the retained empirical eigenpairs with these signal groups. We obtain an orthogonal R , $V = UR$, and a deterministic $\nu_n \rightarrow 0$ such that

$$\|V\|_{2 \rightarrow \infty} \leq C_H/\sqrt{n}, \quad \|V - AU_*\Lambda_*^{-1}\|_{2 \rightarrow \infty} \leq \nu_n/\sqrt{n}.$$

The same alignments satisfy Davis–Kahan's bound $\|V - U_*\|_{\text{op}} \leq C_H/\sqrt{nP_{\max}}$. Set $M = R^\top \Lambda R = V^\top AV$. Since $\|\mathbb{E}[A \mid z]\|_{\text{op}} = O(nP_{\max})$,

$$\|M - \Lambda_*\|_{\text{op}} \leq \|A - \mathbb{E}[A \mid z]\|_{\text{op}} + 2\|\mathbb{E}[A \mid z]\|_{\text{op}}\|V - U_*\|_{\text{op}} \leq C_H\sqrt{nP_{\max}}.$$

The degree union bound from Lemma 3.2 also gives $\max_i \deg(i) \leq C_HnP_{\max}$ at the stated confidence.

Set $E = V - AU_*\Lambda_*^{-1}$. The exact row decomposition is

$$(\tilde{A} - A\Pi_*)_i = E_iMV^\top + (AU_*)_i\{\Lambda_*^{-1}MV^\top - U_*^\top\}.$$

The bracket has operator norm $O((nP_{\max})^{-1/2})$, while $\|(AU_*)_i\|_2 \leq C_H n P_{\max} / \sqrt{n}$. Consequently

$$\|\tilde{A} - A\Pi_*\|_{2 \rightarrow \infty} \leq \frac{C_H}{\sqrt{n}} \{nP_{\max}\nu_n + \sqrt{nP_{\max}}\} = o(nP_{\max}/\sqrt{n}).$$

Cauchy–Schwarz gives the uniform row ℓ_1 bound. For a fixed realization of the random floor ε_n , the clipping map $x \mapsto \max\{x, \varepsilon_n\}$ is 1-Lipschitz. Applying this same map to both reconstructions therefore gives the $q - q^0$ bound without conditioning on the floor. Moreover,

$$|\tilde{A}_{ij}| \leq \|U_i\|_2 \|\Lambda\|_{\text{op}} \|U_j\|_2 \leq C_H P_{\max}.$$

The nonnegative adjacency matrix has spectral radius $\|\Lambda\|_{\text{op}} = \lambda_1(A)$. The row sums of its mean, together with the noise norm bound, give for large n

$$cnP_{\max} \leq \|\Lambda\|_{\text{op}} \leq nP_{\max} + C_H \sqrt{nP_{\max}} = (1 + o(1))nP_{\max}.$$

Hence the admissible adaptive floor satisfies $c'P_{\max} \leq \varepsilon_n = c_0 \max\{\|\Lambda\|_{\text{op}}, 1\}/n \leq P_{\min}/2$ on this event for large n , because $c_0 < c_r/2$. This also proves the common positive box used in the fitting arguments.

For the average approximation, let $N_{ic} = \sum_{j: z_j=c} A_{ij}$. Since $\varepsilon_n < P_{ac}$ on this event, clipping toward P_{ac} cannot increase the coordinate error:

$$\|q_i^0 - p_{z_i,*}\|_1 \leq \sum_c |N_{ic} - n_c P_{z_i,c}|.$$

Let F be the sum on the right over all i . Its expectation is at most $Cn\sqrt{nP_{\max}} + CnP_{\max}$; the last term accounts for the missing diagonal. Changing one independent edge changes F by at most two. The bounded-differences inequality gives

$$\Pr\{F - \mathbb{E}F > n(nP_{\max})^{3/4}\} \leq e^{-c(nP_{\max})^{3/2}}.$$

This is smaller than $e^{-HnP_{\max}}$ for every fixed H and large n . Combining these estimates and enlarging a deterministic vanishing a_n proves every assertion on a common event of probability $1 - O(e^{-HnP_{\max}})$. $\square$

Clipped oracle comparisons with a random floor. Let G_H be the graph event just proved. Fix a true class a and a competitor b , and write $v_c = \log(P_{ac}/P_{bc})$, $N'_{ic} = \max\{N_{ic}, n_c \varepsilon_n\}$, and

$$L_{i,\text{clip}}^{ab} = \sum_c N'_{ic} v_c - \sum_c n_c (P_{ac} - P_{bc}).$$

The floor depends on the whole graph, so its value cannot be conditioned on while asserting independence of the block counts. Instead, on G_H it has the deterministic upper bound $\bar{\varepsilon}_n = P_{\min}/2$. Put $t_c = n_c \bar{\varepsilon}_n$. For large n , $t_c \leq \mathbb{E}_a N_{ic}$, also in the block containing the excluded self-loop. For $\alpha \in [0, 1]$, define the independent count envelopes

$$B_c(N_{ic}) = \begin{cases} e^{-\alpha v_c N_{ic}}, & v_c \geq 0, \\ e^{-\alpha v_c \max\{N_{ic}, t_c\}}, & v_c < 0. \end{cases}$$

There is the pathwise inequality

$$e^{-\alpha L_{i,\text{clip}}^{ab}} \mathbf{1}_{G_H} \leq e^{\alpha \sum_c n_c (P_{ac} - P_{bc})} \prod_c B_c(N_{ic}).$$

When $v_c < 0$, put $\beta = -\alpha v_c \geq 0$. Jensen's inequality and $t_c \leq \mathbb{E}_a N_{ic}$ give

$$\mathbb{E}_a e^{\beta \max\{N_{ic}, t_c\}} \leq \mathbb{E}_a e^{\beta N_{ic}} + e^{\beta t_c} \leq 2\mathbb{E}_a e^{\beta N_{ic}}.$$

The right-hand envelopes have deterministic thresholds, so their moments factor over the independent column-block counts. Thus clipping costs at most 2^K in this restricted Chernoff moment. A direct binomial moment expansion, including the excluded diagonal, gives uniformly in α

$$\log \mathbb{E}_a e^{-\alpha L_{i,\text{raw}}^{ab}} = -D_\alpha(p_{a,-i} \| p_{b,-i}) + O(nP_{\max}^2 + P_{\max}),$$

where the raw score replaces N'_{ic} by N_{ic} . Since $P_{\max} \rightarrow 0$, this remainder is $o(nP_{\max})$. For any deterministic $\tau_n = o(nP_{\max})$, Markov's inequality at the Chernoff maximizer therefore yields

$$\Pr\{L_{i,\text{clip}}^{ab} \leq \tau_n, G_H\} \leq 2^K \exp\{-D_{i,ab}^* + o(nP_{\max})\}. \quad (\text{A.3})$$

This comparison handles the adaptive floor without an independence assumption, and does not require clipping every rare-row score to change it by $o(nP_{\max})$.

Proof of Theorem 3.6. Choose a fixed $H > 2 + \limsup_n I_n/(nP_{\max})$ in Lemma 3.5 and work on G_H . The limsup is finite by the oracle information bounds. Constants below may depend on the model bounds and floor, but not on the iteration. Suppose an aligned responsibility matrix R has average error at most γ , with γ below a sufficiently small fixed constant. Put $N_a = \sum_i R_{ia}$ and $\bar{q}_a = n_a^{-1} \sum_{i:z_i=a} q_i$. Then $|N_a - n_a| \leq n\gamma$ and $N_a \geq \pi_{\min} n/2$. The average-profile bound gives $\|\bar{q}_a - p_{a*}\|_1 \leq C a n P_{\max}$. The M-step identity gives, deterministically,

$$\|\hat{p}_a - \bar{q}_a\|_1 \leq \frac{\sum_i |R_{ia} - Z_{ia}| \|q_i\|_1}{N_a} + \left| \frac{n_a}{N_a} - 1 \right| \|\bar{q}_a\|_1 \leq C \gamma n P_{\max}.$$

Consequently

$$\max_a \|\hat{p}_a - p_{a*}\|_1 \leq C(\gamma + a_n) n P_{\max}, \quad c' P_{\max} \leq \hat{p}_{aj} \leq C_H P_{\max}. \quad (\text{A.4})$$

The logarithm is $C/P_{\max}$ -Lipschitz on the common positive box. Since $q_{ij} \leq C_H P_{\max}$, fitted linear contrasts and offsets therefore differ from their true-profile working counterparts by at most $C(\gamma + a_n) n P_{\max}$. The fitted log-weight ratios are bounded: the unconstrained weights N_a/n are bounded away from zero; a fixed positive weight floor gives the same bound directly.

Combining the uniform $q - q^0$ bound with these parameter bounds gives, for every node and pair,

$$|\hat{\ell}_i(a) - \hat{\ell}_i(b) - L_{i,\text{clip}}^{ab}| \leq C(\gamma + a_n) n P_{\max} + C. \quad (\text{A.5})$$

No independence of R , q , and the incident edges is used. For $\gamma = \gamma_n \rightarrow 0$, a wrong classification on the weak-start and graph events implies $L_{i,\text{clip}}^{ab} \leq o(nP_{\max})$ for a competitor. Equation (A.3) and a finite union bound prove the one-update rate after adding the two exceptional-event probabilities.

For further updates, the conditioning assumption and the positive box give a fixed $\kappa > 0$ with $\min_{a \neq b} D(p_{a*} \| p_{b*}) \geq \kappa n P_{\max}$ for large n . Choose a fixed basin radius $\gamma_0 > 0$ so that the fitted-score error for $\gamma \leq \gamma_0$ is less than $\kappa n P_{\max}/8$ in each comparison. Choose a fixed small $t_0 > 0$ so that $\|q_i - p_{z_i,*}\|_1 \leq t_0 n P_{\max}$ costs less than $\kappa n P_{\max}/8$ in the observation contrast. At $q_i = p_{z_i,*}$, the population working margin is $D(p_{z_i,*} \| p_{b*}) \geq \kappa n P_{\max}$. After the observation, parameter, and bounded weight corrections, each specified row retains a correct-class margin of at least $\kappa n P_{\max}/4$

for large n . The average-profile bound leaves at most a_n/t_0 exceptional rows. The next E-step thus has aligned average responsibility error at most

$$b_n = 2a_n/t_0 + 2(K-1)e^{-\kappa n P_{\max}/4} \rightarrow 0.$$

This holds uniformly throughout the fixed basin. Eventually $b_n < \gamma_0$, so the basin is invariant. Starting from $\gamma_n \rightarrow 0$, every later responsibility matrix has error at most $\max\{\gamma_n, b_n\} = o(1)$. Equation (A.5) then has one deterministic $o(nP_{\max})$ bound for every later score on the same event, including a data-dependent finite stopping time. No union bound over iterations is needed.

Finally, add the initialization failure δ_n and the graph failure $O(e^{-Hn P_{\max}})$ to the per-node Chernoff bound and average over nodes. The chosen H makes the graph failure negligible relative to e^{-I_n} , and the initialization reliability preserves that exponent. If $\liminf I_n/\log n > 1$, multiplying the expected error by n and applying Markov's inequality proves exact recovery. $\square$

A.6 End-to-end likelihood initialization

Theorem A.2 (Original growing branch for two communities). *Under (2.2)–(2.3) and the admissible observable floor, the global-mean growing branch alone for $K = 2$, followed by at least one exact profile and weight M -step, satisfies $\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}$. The same conclusion holds at any subsequent finite exact-EM stopping time, and exact recovery holds if $\liminf I_n/\log n > 1$.*

Proof of Theorem A.2. Fix $H > 2 + \limsup_{n \rightarrow \infty} I_n/(nP_{\max})$ and work on the event of Lemma 3.5. All constants below may depend on H and the fixed model. Enlarge its deterministic $a_n \rightarrow 0$ if necessary to bound both displayed profile errors. Put $\pi_{a,n} = n_a/n$,

$$m_b = \sum_{a=1}^2 \pi_{a,n} P_{ab}, \quad D_{\pi_n}(x||y) = \sum_b \pi_{b,n} \{x_b \log(x_b/y_b) - x_b + y_b\}.$$

The block values of q_c^0 are

$$v_{cb} = \max \left\{ \frac{\sum_{j: z_j=b} A_{cj}}{n_b}, \varepsilon_n \right\}, \quad q_{cj}^0 = v_{c,z_j}.$$

This is only an analysis variable. In particular v_c may be a rare extreme candidate, and we do not assume it is near its own community profile. Let

$$h_a(v) = \frac{1}{P_{\max}} \sum_b \pi_{b,n} \{P_{ab} \log(v_b/m_b) - (v_b - m_b)\}, \quad \Phi_n(v) = \sum_a \pi_{a,n} [h_a(v)]_+.$$

The global profile mean $\bar{q}$ differs from $(m_{z_j})_{j=1}^n$ in ℓ_1 norm by at most $a_n n P_{\max}$. The uniform bound on $q_c - q_c^0$ and the common positive box give

$$|[\ell_i(q_c) - \ell_i(\bar{q})] - n P_{\max} h_{z_i}(v_c)| \leq C\{\|q_i - p_{z_i,*}\|_1 + a_n n P_{\max}\}.$$

Here the parameter perturbation costs $C\|q_c - q_c^0\|_1 + C\|\bar{q} - (m_{z_j})_{j=1}^n\|_1$, because $q_{ij} \leq C P_{\max}$ and log is $C/P_{\max}$ -Lipschitz. The remaining observation perturbation multiplies a uniformly bounded log ratio. Since positive part is 1-Lipschitz, summing yields the simultaneous candidate bound

$$\sup_c \left| \frac{\Delta_2(c)}{n^2 P_{\max}} - \Phi_n(v_c) \right| \leq C a_n.$$

Every true class has a candidate with $\|q_c - p_{a*}\|_1 = o(nP_{\max})$ by the average error bound and its positive proportion. The same is true of q_c^0 , so $P_{\max}^{-1} \sum_b \pi_{b,n} |v_{cb} - P_{ab}| = o(1)$. The uniform conditioning and positive community proportions imply that

$$g_0 = \liminf_{n \rightarrow \infty} \max_a \frac{\pi_{a,n}}{P_{\max}} D_{\pi_n}((P_{ab})_{b=1}^2 \|m) > 0.$$

Positivity follows because the conditioned block rows are uniformly separated relative to $P_{\max}$ and cannot both approach their mean at that scale. Uniform Lipschitz continuity after dividing all block values by $P_{\max}$ and the candidate bound imply that the selected candidate satisfies $\Phi_n(v_c) \geq g_0/2$ for large n .

The decisive two-class identity is

$$\sum_a \pi_{a,n} h_a(v) = -\frac{1}{P_{\max}} D_{\pi_n}(m \| v) \leq 0.$$

Together with $\Phi_n(v_c) \geq g_0/2$, it forces exactly one of $h_1(v_c), h_2(v_c)$ to be positive. Moreover

$$h_{\text{pos}}(v_c) \geq g_0/2, \quad h_{\text{neg}}(v_c) \leq -g_0/2,$$

since each weighted positive or negative magnitude is at least $g_0/2$ and each proportion is at most one. The initial two mixture weights are equal, so they do not alter this score comparison.

Take $\tau_n = \sqrt{a_n} + (nP_{\max})^{-1}$. At most $(a_n/\tau_n)n = o(n)$ observations have $\|q_i - p_{z_i,*}\|_1 > \tau_n n P_{\max}$. Every other observation has the correct initial component margin at least $g_0 n P_{\max}/4$ after the above score perturbations. After a common label alignment, the first E-step therefore satisfies

$$\frac{1}{n} \sum_{i,a} |R_{ia} - Z_{ia}| \leq \frac{2a_n}{\tau_n} + 2e^{-g_0 n P_{\max}/4} = o(1).$$

The graph event has failure $O(e^{-Hn P_{\max}})$, so these responsibilities meet the reliability condition in Theorem 3.6. The evaluated fitting loop always performs its first M-step before it may stop. That theorem gives the claimed exponent for the first resulting decode and all subsequent finite updates, including the evaluated stopping rule. Its exact-recovery implication also applies. $\square$

A.7 Deterministic likelihood replacement for every fixed K

Lemma A.3 (Iterative likelihood replacement certifies a weak start). *Let $K \geq 2$ be fixed, put $P_{\max} = \max_{a,b} P_{ab}$, and suppose $nP_{\max} \rightarrow \infty$. Assume, for constants $0 < c \leq C < \infty$, $\pi_{\min} > 0$, and $s > 0$, that*

$$n_a \geq \pi_{\min} n, \quad cP_{\max} \leq q_{ij}, p_{aj} \leq CP_{\max}, \quad (\text{A.6})$$

$$\sum_i \|q_i - p_{z_i,*}\|_1 \leq a_n n^2 P_{\max}, \quad a_n \rightarrow 0, \quad (\text{A.7})$$

$$\min_{a \neq b} \|p_{a*} - p_{b*}\|_1 \geq snP_{\max}. \quad (\text{A.8})$$

Start at any K fitted profiles in the convex hull of the rows of q and any feasible simplex weights. A fixed weight floor $f \in [0, 1/K)$ is allowed. Define

$$\begin{aligned} D(x\|y) &= \sum_j \{x_j \log(x_j/y_j) - x_j + y_j\}, \\ J(\hat{p}) &= \sum_i \min_{k \leq K} D(q_i\|\hat{p}_{k*}), \end{aligned} \quad (\text{A.9})$$

$$F(\hat{\pi}, \hat{p}) = - \sum_i \log \sum_{k=1}^K \hat{\pi}_k e^{-D(q_i\|\hat{p}_{k*})}, \quad (\text{A.10})$$

$$T_n = \sum_{i,j} q_{ij}, \quad \zeta_n = (\log n)^{-1/2}, \quad M_n = \lceil (\log n)^2 \rceil. \quad (\text{A.11})$$

At each replacement round form the following $K+1$ proposals. One proposal starts at the current profiles with weights $1/K$. For each component r , retain the other $K-1$ profiles and choose

$$c_r \in \arg \max_{c \in [n]} \sum_i \left(\ell_i(q_c) - \max_{k \neq r} \ell_i(\hat{p}_{k*}) \right)_+. \quad (\text{A.12})$$

Replace profile r by q_{c_r} and initialize all K weights at $1/K$. Fit each proposal by any prescribed finite number of exact EM updates, including zero updates, whose M -steps maximize over the stated weight simplex. A zero-mass profile retains its previous value. Select the endpoint with largest $\mathcal{L}$ and accept it only when its objective increase is at least $\zeta_n T_n$; otherwise stop. After M_n accepted rounds, return the current fit without further replacement. Candidate scans in (A.12) include all node IDs, even those used in previous rounds. All ties are resolved deterministically.

For all sufficiently large n , the cap M_n is not reached, and the algorithm stops after at most $C_1 \sqrt{\log n}$ accepted replacements, where C_1 depends only on the constants in (A.6). At its endpoint, put

$$b_n = a_n + \zeta_n + \frac{\log K}{nP_{\max}} \longrightarrow 0.$$

There is a constant C_2 , depending only on the constants in the assumptions and K , such that

$$\max_a \min_k D(p_{a*}\|\hat{p}_{k*}) \leq C_2 b_n n P_{\max}, \quad (\text{A.13})$$

$$J(\hat{p}), F(\hat{\pi}, \hat{p}) \leq C_2 b_n n^2 P_{\max}. \quad (\text{A.14})$$

After one common permutation of the fitted components,

$$\max_a \|\hat{p}_{a*} - p_{a*}\|_1 \leq C_2 \sqrt{b_n} n P_{\max}. \quad (\text{A.15})$$

For the endpoint's exact E-step responsibilities R , the same permutation gives a deterministic $\gamma_n \rightarrow 0$ such that

$$\frac{1}{n} \sum_{i,a} |R_{ia} - Z_{ia}| \leq \gamma_n. \quad (\text{A.16})$$

The statement is uniform over the starting fit, all selected candidates, and the finite EM stopping rules.

Proof. Write $\mathcal{B}_n = [cP_{\max}, CP_{\max}]^n$ and enlarge C if necessary without changing c . All starting and proposed profiles lie in $\mathcal{B}_n$. Each profile M -step is a convex combination of rows of q , and retaining the previous profile in a zero-mass component preserves this property. Thus the same box contains

every profile at every replacement round. Uniform weights are feasible because $f < 1/K$. Exact EM never decreases $\mathcal{L}$, including with the floored simplex, by Proposition B.3.

Score identities and bounds uniform over fitted profiles. The scalar derivatives of $D(x\|y)$ are $\log(x/y)$ and $1 - x/y$. They are uniformly bounded on the common box. Consequently, for a constant L independent of n and all $x, y, x', y' \in \mathcal{B}_n$,

$$0 \leq D(x\|y) \leq LnP_{\max}, \quad (\text{A.17})$$

$$|D(x\|y) - D(x'\|y')| \leq L\{\|x - x'\|_1 + \|y - y'\|_1\}. \quad (\text{A.18})$$

These bounds apply to random fitted profiles and extreme candidate rows; they make no assumption that a selected candidate is close to its true class profile. Define the observation-only constant

$$S_q = \sum_{i,j} (q_{ij} \log q_{ij} - q_{ij}).$$

Since $\ell_i(\hat{p}_{k*}) = \ell_i(q_i) - D(q_i\|\hat{p}_{k*})$, the working objective satisfies the exact identities and inequalities

$$\mathcal{L} = S_q - F, \quad 0 \leq J \leq F, \quad F(\mathbf{1}/K, \hat{p}) \leq J(\hat{p}) + n \log K. \quad (\text{A.19})$$

For example, the last inequality follows by retaining a minimizing term in each row's sum. It does not compare weights from different fits by an unstated monotonicity assumption.

Introduce the population hard cost

$$\bar{J}(\hat{p}) = \sum_{a=1}^K n_a \min_k D(p_{a*}\|\hat{p}_{k*}).$$

For any fixed K -tuple $\hat{p} \in \mathcal{B}_n^K$, the elementary inequality $|\min_k u_k - \min_k v_k| \leq \max_k |u_k - v_k|$ and (A.18) imply

$$\sup_{\hat{p} \in \mathcal{B}_n^K} |J(\hat{p}) - \bar{J}(\hat{p})| \leq La_n n^2 P_{\max}. \quad (\text{A.20})$$

The supremum is the reason that adaptively selected centers and data-dependent stopping do not require additional probability bounds. For every true class a , averaging (A.7) over its at least $\pi_{\min} n$ nodes supplies a candidate $c(a)$ in that class with

$$\|q_{c(a)} - p_{a*}\|_1 \leq (a_n/\pi_{\min}) n P_{\max}. \quad (\text{A.21})$$

These candidates are used only in the proof; the algorithm does not know their labels.

A population replacement supplied by pigeonhole. Fix the current K centers. Assign each of the K true profiles to a minimizing center, using any deterministic tie rule, and write $k(a)$ for this assignment and $v_a = D(p_{a*}\|\hat{p}_{k(a),*})$. If $k(a)$ is assigned no true profile other than a , remove $k(a)$. If $k(a)$ is assigned at least two true profiles, at most $K - 1$ centers can have any assigned profile, so choose an unassigned center to remove. In either case, every true class $b \neq a$ retains its assigned minimizing center. Inserting the exact profile p_{a*} would therefore lower $\bar{J}$ by at least $n_a v_a$, because the new loss of class a is zero and the losses of the other classes cannot increase. Inserting the actual row $q_{c(a)}$ instead, (A.21) and (A.18) give a replacement configuration $\hat{p}^{\text{trial}}$ satisfying

$$\bar{J}(\hat{p}^{\text{trial}}) \leq \bar{J}(\hat{p}) - n_a v_a + L\pi_{\min}^{-1} a_n n^2 P_{\max}. \quad (\text{A.22})$$

Here the inserted profile may benefit other classes; such benefits are unneeded for the inequality. Exact score ties cause no difficulty because the retained minimizing center is still available to every other class.

The implemented gain finds a replacement at least as good. For a removed component r , write

$$J_{-r} = \sum_i \min_{k \neq r} D(q_i \| \hat{p}_{k*}), \quad h_i^{-r} = \max_{k \neq r} \ell_i(\hat{p}_{k*}).$$

Directly from the score identity,

$$J(\hat{p}^{(r,c)}) = J_{-r} - \sum_i (\ell_i(q_c) - h_i^{-r})_+, \quad (\text{A.23})$$

where $\hat{p}^{(r,c)}$ replaces r by q_c . Thus (A.12) is an exact minimization of the post-replacement hard cost for that deletion. In particular the candidate chosen for the deletion in (A.22) has empirical hard cost no larger than the proof candidate's cost. Applying (A.20) to both configurations gives, for a constant L_1 ,

$$J(\hat{p}^{(r,c_r)}) \leq J(\hat{p}) - n_a v_a + L_1 a_n n^2 P_{\max}.$$

Uniform proposal weights and (A.19), followed by monotonicity during its fitting, imply that its endpoint has

$$F_{\text{proposal}} \leq F_{\text{current}} - n_a v_a + L_1 a_n n^2 P_{\max} + n \log K. \quad (\text{A.24})$$

This inequality holds for every a and its indicated deletion, so the best evaluated endpoint is at least as good. No purity, convergence, or basin assumption on the growing initialization has been used.

The cap is inactive and stopping bounds every population loss. The common box gives $cn^2 P_{\max} \leq T_n \leq Cn^2 P_{\max}$. For any feasible weights, (A.17) gives $0 \leq F \leq Ln^2 P_{\max}$, because every exponential in each row's weighted sum is at least $e^{-Ln P_{\max}}$. Each accepted replacement decreases F by at least $\zeta_n T_n$. There are therefore at most $L/(c\zeta_n)$ accepted rounds. Since this is $O(\sqrt{\log n}) < M_n$ eventually, the prescribed cap is inactive for all sufficiently large n under the stated deterministic assumptions. The algorithm must terminate by rejection of all $K + 1$ proposals.

At that endpoint, (A.24) and rejection give

$$n_a v_a \leq L_1 a_n n^2 P_{\max} + n \log K + \zeta_n T_n.$$

Dividing by $n_a \geq \pi_{\min} n$ proves (A.13). Summing over a and using (A.20) proves the asserted bound on J . The rejected current-centers uniform-weight proposal has endpoint cost at most $J + n \log K$, so its rejection also implies

$$F_{\text{current}} \leq J + n \log K + \zeta_n T_n.$$

This proves the bound on F in (A.14). The additional proposal is needed: small current weights can otherwise make F large even when J and all center errors are small.

A single center matching and weak responsibilities. The Hessian of $\sum_j x_j \log x_j - x_j$ is $\text{diag}(1/x_j)$. Strong convexity on $\mathcal{B}_n$, and then Cauchy–Schwarz, give

$$D(x \| y) \geq \frac{\|x - y\|_2^2}{2CP_{\max}} \geq \frac{\|x - y\|_1^2}{2CnP_{\max}}. \quad (\text{A.25})$$

By (A.13), each true class has a fitted center within $C_3 \sqrt{b_n} n P_{\max}$ in ℓ_1 . If two distinct classes chose the same center, their mutual distance would be at most $2C_3 \sqrt{b_n} n P_{\max}$, contrary to (A.8) for large n . With exactly K true profiles and K fitted profiles, this injection is a bijection. It supplies the common permutation in (A.15).

Set $t_n = \sqrt{a_n} + (nP_{\max})^{-1}$ and call row i typical when $\|q_i - p_{z_i,*}\|_1 \leq t_n n P_{\max}$. By (A.7), the fraction of nontypical rows is at most $a_n/t_n \rightarrow 0$. For a typical row in class a and any incorrectly matched component $k \neq a$, separation and the center matching imply

$$\|q_i - \hat{p}_{k*}\|_1 \geq (s/2)n P_{\max}$$

for all sufficiently large n . By (A.25), its deviance is at least $c_* n P_{\max}$, with $c_* = s^2/(8C) > 0$. The exact E-step satisfies

$$R_{ik} = \frac{\hat{\pi}_k e^{-D(q_i \| \hat{p}_{k*})}}{\sum_l \hat{\pi}_l e^{-D(q_i \| \hat{p}_{l*})}}.$$

Its row-wise variational identity is

$$-\log \sum_k \hat{\pi}_k e^{-D(q_i \| \hat{p}_{k*})} = \sum_k R_{ik} D(q_i \| \hat{p}_{k*}) + \text{KL}(R_i \| \hat{\pi}) \geq \sum_k R_{ik} D(q_i \| \hat{p}_{k*}).$$

Summing over typical rows and using $\sum_k |R_{ik} - Z_{ik}| = 2 \sum_{k \neq z_i} R_{ik} \leq 2$ gives

$$\frac{1}{n} \sum_{i,k} |R_{ik} - Z_{ik}| \leq \frac{2a_n}{t_n} + \frac{2F}{c_* n^2 P_{\max}} \leq \frac{2a_n}{t_n} + \frac{2C_2 b_n}{c_*} =: \gamma_n \rightarrow 0.$$

The identities remain valid for zero weights with the usual convention that the corresponding responsibilities vanish. In particular, no unstated lower bound on the endpoint's fitted mixture weights is required for this weak-start argument. All bounds are simultaneous for every configuration in the box, completing the deterministic proof. $\square$

Proof of Theorem 3.7. Fix a confidence constant H sufficiently large that the exceptional probability in Lemma 3.5 is negligible relative to e^{-I_n} . Under the density-uniform form of that lemma, its probability is $O(e^{-HnP_{\max}})$, and the upper bound $I_n \leq C_I n P_{\max}$ allows any fixed $H > C_I$. Work on its spectral-profile event. The positive probability ratios, positive limiting community proportions, and the spectral conditioning assumption give (A.6) and (A.8), with constants that may depend on H . Its average-profile bound is exactly (A.7). The growing fit starts at a row mean and only appends rows and takes convex profile updates, so all of its endpoint profiles lie in the required row convex hull.

Apply Lemma A.3 to the deterministic replacement stage of Algorithm 1. For every sufficiently large n on this event, its round cap is inactive and its endpoint E-step obeys (A.16). Thus the local initialization condition (3.18) holds with a deterministic $\gamma_n \rightarrow 0$ and $\delta_n = O(e^{-HnP_{\max}})$. There is no additional randomized-initialization failure. The mandatory final profile and weight M-step and Theorem 3.6 give

$$\mathbb{E} \text{Mis}(\hat{z}, z) \leq e^{-(1+o(1))I_n}.$$

The same theorem covers any further exact updates and their stated finite stopping rules. If $\liminf I_n / \log n > 1$, multiplying the expected error fraction by n and applying Markov's inequality proves exact recovery with probability tending to one. All replacement proposals use only q , hence only the retained eigenpairs, and the number of replacement rounds and the fitting iteration budgets are polynomial by construction. No recovery property of the unrepaired growing endpoint is needed. $\square$

B Proofs of initialization and optimization properties

B.1 Frozen-candidate greedy gains

Proposition B.1 (Greedy gain guarantee). *Let s_{ic} be fixed finite candidate scores and let S_0 be a fixed nonempty seed set. For candidate sets T , define*

$$G(T) = \sum_i \max_{c \in S_0 \cup T} s_{ic} - \sum_i \max_{c \in S_0} s_{ic}.$$

Then G is normalized, monotone, and submodular. If $m \geq 1$ greedy steps select a set T_m and T^* maximizes G over sets of at most m candidates, then

$$G(T_m) \geq \{1 - (1 - 1/m)^m\}G(T^*) \geq (1 - e^{-1})G(T^*).$$

Proof. Normalization and monotonicity follow immediately from the definition. The marginal gain of candidate c is

$$G(T \cup \{c\}) - G(T) = \sum_i \left(s_{ic} - \max_{b \in S_0 \cup T} s_{ib} \right)_+.$$

If $T \subseteq T'$, each maximum in the right side increases, so the marginal gain decreases. This proves submodularity. For the greedy set T_t at time t , monotonicity and telescoping diminishing returns give

$$\begin{aligned} G(T^*) - G(T_t) &\leq G(T_t \cup T^*) - G(T_t) \\ &\leq \sum_{c \in T^* \setminus T_t} \{G(T_t \cup \{c\}) - G(T_t)\} \\ &\leq m\{G(T_{t+1}) - G(T_t)\}. \end{aligned}$$

Starting from $G(T_0) = 0$ and iterating this inequality proves the claimed factor. The final inequality follows from $(1 - 1/m)^m \leq e^{-1}$, including the case $m = 1$. $\square$

This is the standard cardinality-constrained submodular greedy guarantee [7]. It controls added explanation relative to the fixed initial seed, not the multiplicative approximation of a possibly negative total log score. It does not apply unchanged when intermediate EM steps alter the candidate scores.

B.2 Noiseless profile selection

Proposition B.2 (Population profile selection). *Suppose $q_i = \theta_{z_i}$, where the K positive profile vectors $\theta_1, \dots, \theta_K$ are pairwise distinct and every class is nonempty. The candidate pool consists of these node profiles. From any first seed belonging to this candidate pool, $K - 1$ subsequent global-gain greedy steps select all K different profiles. Maximum working-score classification using those profiles recovers z up to permutation.*

Proof. For $x, \hat{p} > 0$ coordinatewise, write $\ell_x(\hat{p}) = \sum_j x_j \log \hat{p}_j - \sum_j \hat{p}_j$. Then

$$\ell_x(x) - \ell_x(\hat{p}) = \sum_j \{x_j \log(x_j/\hat{p}_j) - x_j + \hat{p}_j\} = D(x\|\hat{p}) \geq 0,$$

with equality precisely when $x = \hat{p}$; the inequality follows from $\log u \leq u - 1$. If a profile has already been selected, selecting a duplicate has zero gain. If profile θ_b has not been selected, then each node in class b gains $\min_{a \in S} D(\theta_b\|\theta_a) > 0$ when it is added. Therefore some unselected profile has strictly positive gain, and greedy cannot select a duplicate before all distinct profiles have been selected. The same strict divergence inequality makes each node's own profile its unique best score after selection. $\square$

The proposition concerns the frozen node-profile selection control in an abstract noiseless profile model. It does not directly cover Algorithm 1, whose initial mean need not be a node profile and whose intermediate EM fits change the parameters. It does not replace a stability analysis for random spectral profiles, and it does not erase the diagonal distinction in a no-self-loop SBM.

B.3 Exact working EM and accepted replacements

Proposition B.3 (Objective convergence of exact working EM). *For fixed strictly positive profiles $q_1, \dots, q_n$, let*

$$\mathcal{L}(\hat{\pi}, \hat{p}) = \sum_i \log \sum_{a=1}^K \hat{\pi}_a \exp\{\ell_i(\hat{p}_a)\}, \quad \ell_i(\hat{p}_a) = \sum_j q_{ij} \log \hat{p}_{aj} - \sum_j \hat{p}_{aj}.$$

Starting with positive parameters and positive mixture weights, the exact E-step and unconstrained exact M-step

$$r_{ia} = \frac{\hat{\pi}_a e^{\ell_i(\hat{p}_a)}}{\sum_b \hat{\pi}_b e^{\ell_i(\hat{p}_b)}}, \quad N_a = \sum_i r_{ia},$$

$$\hat{\pi}_a^{\text{new}} = N_a/n, \quad \hat{p}_{aj}^{\text{new}} = \frac{\sum_i r_{ia} q_{ij}}{N_a}$$

make $\mathcal{L}$ nondecreasing. Its numerical values converge to a finite limit. The same monotonicity statement holds for an exact M-step maximizing over any stated constraint set containing the current parameters, and for replacement proposals accepted only when their final $\mathcal{L}$ improves.

Proof. For row-wise probability vectors r_i , define the variational objective

$$\mathcal{F}(r, \hat{\pi}, \hat{p}) = \sum_{i,a} r_{ia} \{\log \hat{\pi}_a + \ell_i(\hat{p}_a) - \log r_{ia}\}.$$

The log-sum inequality gives $\mathcal{F}(r, \hat{\pi}, \hat{p}) \leq \mathcal{L}(\hat{\pi}, \hat{p})$, with equality for the E-step responsibilities. For fixed r , maximizing $\sum_a N_a \log \hat{\pi}_a$ over the simplex gives $\hat{\pi}_a = N_a/n$. For each profile coordinate, maximizing $(\sum_i r_{ia} q_{ij}) \log \hat{p}_{aj} - N_a \hat{p}_{aj}$ gives the displayed weighted mean. These are interior maxima because all inputs and responsibilities are positive. Thus the E-step makes the variational bound tight and the M-step raises it, proving monotonicity.

The saturated-score inequality used in the preceding proof implies $\ell_i(\hat{p}_a) \leq \ell_i(q_i)$ for every i, a . Since $\sum_a \hat{\pi}_a = 1$,

$$\log \sum_a \hat{\pi}_a e^{\ell_i(\hat{p}_a)} \leq \ell_i(q_i).$$

Consequently $\mathcal{L} \leq \sum_i \ell_i(q_i) < \infty$. A nondecreasing real sequence bounded above has a finite limit. For a constrained exact M-step the current parameters remain feasible, so the same variational argument applies. The replacement assertion follows directly from its acceptance rule. $\square$

This proposition proves convergence of objective values. It proves neither convergence to the global maximum nor recovery of the true labels. The implementation uses an exact active-set maximizer on the floored simplex, so its weight constraint is covered by the constrained version of this proposition. Its finite stopping tolerance and floating-point accuracy are assessed separately in Section 4.

Proposition B.4 (Profile hull and spectral dimension). *Every responsibility-weighted profile M-step produces a profile in the convex hull of the rows of q . If the unmodified reconstruction $q = U\Lambda U^\top$ is used and the resulting means are positive, all fitted profiles lie in its row space, of dimension at most K . Entrywise positive clipping can increase this dimension.*

Proof. The coefficients R_{ia}/N_a are nonnegative and sum to one, so the update is a row convex combination. Every such combination also lies in the row space. The unmodified reconstruction has rank at most K . The clipping operation is nonlinear and has no corresponding rank-preservation identity. $\square$

C Implementation and reproducibility

C.1 Experimental entry points and retained outputs

The numerical evidence is archived in the repository

<https://github.com/zhixin0825/spectral-lrt>.

Algorithm 1 is implemented in `residual_seed.py`. Its method is

`Work.iterative_certified_growing()`.

The public API defaults to `method='global_gain_certified'`. It starts with the global-mean growing fit and then iterates deterministic component replacement. Each round fits the current profiles from uniform weights and, for each component removal, fits the globally selected replacement from uniform weights. Every node ID participates in each replacement scan. The largest trial objective is accepted only if its improvement is at least $T/\sqrt{\log n}$, where $T = \sum_{ij} q_{ij}$. The accepted-round cap is $\lceil (\log n)^2 \rceil$, followed by one mandatory final E/M update. Under the theorem's hypotheses, the stopping certificate is reached in $O(\sqrt{\log n})$ accepted rounds, before the cap. The earlier residual-restart implementation is exposed separately as `method='global_gain_restarts'`. The archived primary benchmark remains the original growing branch. The principal comparison is in `experiments/20261009_residual_likelihood_seeding/`. Its report, protocol, model-generation script, fitting code, scalar verification, and per-graph summaries define the numerical experiment. The preceding partial-peeling experiment and its diagnosis are in `experiments/20261009_partial_likelihood_peeling/`. The original growing branch corresponds to the recorded `growing_global_gain_EM` arm and the API option `method='global_gain'`. It starts at the global profile mean, adds $K - 1$ node candidates, and refits after every addition. The `global_gain_no_update` arm is a separate frozen-profile control with a top-subset first seed. Identical final error counts do not identify the algorithms.

The manuscript's two CSV summaries retain batch-specific, combined, and model-specific results. The diagnostic figure uses saved coordinates and fitted labels rather than a new simulation. Full spectral arrays and objective traces are stored in the previously archived numerical records; the source reports describe their restoration. Reconstructing the profiles from (U, Λ) and fitting the working score does not need the original adjacency matrix.

C.2 Logarithm and weight floors

The current implementation sets

$$q_{ij} = \max\{(U\Lambda U^\top)_{ij}, \varepsilon_n\}, \quad \varepsilon_n = 10^{-4} \frac{\max\{\|\Lambda\|_{\text{op}}, 1\}}{n},$$

and uses the exact weight update with floor $f = 10^{-8}$. The spectral scale is computed from the retained eigenvalues. The theorem requires the chosen fixed floor constant to be sufficiently small

relative to the block-probability comparison constants. On the spectral approximation event, this gives a floor of order $P_{\max}$ strictly below every P_{ab} . The value 10^{-4} does not supply such a guarantee for arbitrarily small $P_{\min}/P_{\max}$. This operation defines positive logarithm arguments after spectral reconstruction. Profiles are positive vectors, and finite-sample reconstructed coordinates are not constrained to lie below one.

The archived 240-graph results used the earlier floor $10^{-4} \log n/n$. Those recorded runs and their summaries are retained as historical evidence. Their settings should not be substituted for the adaptive floor of Algorithm 1 when testing densities above logarithmic order.

The API default `method='global_gain_certified'` executes Algorithm 1. The historical replacement branch is `method='repair'`, the original growing branch is `method='global_gain'`, and the earlier random-restart combination is `method='global_gain_restarts'`. These options implement different procedures.

C.3 Historical subset controls

These controls are retained for reproducibility of earlier experiments and are not steps of Algorithm 1. In the frozen global-gain control, the first candidate maximizes the sum of its largest $m_0 = \min\{\lfloor 2n/(3K) \rfloor, n - K\}$ scores over other nodes, excluding itself. In peeling, when M candidates and r components remain, the selected subset contains $\min\{\lfloor 2M/(3K) \rfloor, M - r\}$ other nodes. Only these historical controls require a subset-size rule; the primary growing algorithm uses the global mean and all-node gains throughout.

C.4 Floored-simplex weight updates

For a fixed floor $f \in [0, 1/K)$, the weight M-step solves

$$\max_{\hat{\pi}_a \geq f, \sum_a \hat{\pi}_a = 1} \sum_a N_a \log \hat{\pi}_a. \quad (\text{C.1})$$

Its active-set solution starts with every component free. If the free set is S , assign mass $1 - f(K - |S|)$ proportionally to the N_a , $a \in S$, and fix any weight below f at the floor. Repeat on the remaining free set. The KKT conditions give

$$\hat{\pi}_a = \max\{f, N_a/\lambda\}, \quad \sum_a \hat{\pi}_a = 1 \quad (\text{C.2})$$

for a suitable multiplier λ . The active-set procedure enforces this equation; it is not a naive clipping followed by renormalization. The tested function `_floored_weights` implements this exact constrained update. Since the E-step and mean M-step are unchanged, the variational proof applies on the floored-simplex parameter domain.

C.5 Archived stopping rule

The 240-graph profile fits stop only after hard labels are unchanged, maximum relative mean change and absolute weight change are at most 10^{-3} , and the objective increment is at most $10^{-6}(1 + |\mathcal{L} - G|)$ in absolute value. Here $G = \sum_{ij} \log \Gamma(q_{ij} + 1)$ is a cached observation-only constant used solely to scale the numerical stopping tolerance. It does not enter scoring, responsibilities, M-steps or fit selection. The update cap is 300. These historical numerical choices were preserved in this revision.

C.6 An explicit error decomposition

For fitted components matched to labels a, b , put

$$\hat{w}_j^{ab} = \log(\hat{p}_{aj}/\hat{p}_{bj}), \quad \hat{c}^{ab} = - \sum_j (\hat{p}_{aj} - \hat{p}_{bj}) + \log(\hat{\pi}_a/\hat{\pi}_b). \quad (\text{C.3})$$

If λ^{ab} is the chosen oracle prior log ratio, the empirical comparison has the decomposition

$$\begin{aligned} \hat{T}_i^{ab} - (L_i^{ab} + \lambda^{ab}) &= A_i(\Pi - \text{Id})w^{ab} + (q_i - \tilde{A}_i)w^{ab} \\ &\quad + q_i(\hat{w}^{ab} - w^{ab}) + \hat{c}^{ab} - c_i^{ab} - \lambda^{ab}. \end{aligned} \quad (\text{C.4})$$

The terms correspond to empirical projection, positivity clipping, estimated contrasts, and intercept calibration. The reconstructed diagonal is included in these terms; the oracle excludes it through $A_{ii} = 0$ and the definition of c_i^{ab} . If the exact Bernoulli contrast is replaced by the first-order sparse contrast, Lemma 3.2 accounts for the additional Taylor remainder. Formula (C.4) is exact and gives concrete quantities for checking the transfer conditions. Theorem 3.6 controls the empirical fitting stage through a comparison with clipped oracle block-count scores. It does not require a separate rare-event $o(nP_{\max})$ bound on the raw clipping term in this decomposition. Algorithm 1 has an end-to-end proof for every fixed $K \geq 2$ under the stated minimum-degree, sparsity and signal conditions. Its deterministic repeated-replacement certificate supplies the local theorem’s initialization requirement.

C.7 Verification of the manuscript

The source comprises five numbered main sections, proof appendices, and a bibliography. A build runs PDFLaTeX, BibTeX, and two additional PDFLaTeX passes, checking unresolved citations and references. Build provenance records the source commit and the checksums of the source and compiled PDF. Compilation checks syntax and cross references; mathematical conclusions are exactly those stated with their hypotheses in the text.

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